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Backward Conditioning Reveals Flexibility in Infralimbic Cortex Inhibitory Memories

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eLife Assessment

This set of experiments provides **important** knowledge for how the infralimbic cortex is recruited to inhibit behavior after extinction training. The evidence supporting the conclusions is **convincing** with multiple sophisticated behavioral designs providing converging lines of evidence, though reviewers note possible alternative interpretations and limitations of small group sizes in some cases. This work will be of interest to those interested in cortical function, learning and memory, aversive behavior, and/or related psychiatric factors.

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Abstract

While the infralimbic cortex (IL) is recognized as critical for behavioral inhibition, the content of the inhibitory memories stored in this region remains elusive. To probe this content, we examined some of the conditions that allow retrieval and facilitation of an inhibitory memory stored in the IL using optogenetic stimulation in female and male rats. We found that IL stimulation did not facilitate an initial fear extinction session. However, prior experience with fear extinction enabled IL stimulation to facilitate subsequent fear extinction. Importantly, the facilitative effects of IL stimulation were not limited to fear extinction experience, as prior exposure to backward fear conditioning also enabled IL stimulation to enhance later fear extinction. The effects were stimulus-specific and did not depend on the motivational context present during the prior experience, as backward appetitive conditioning allowed IL stimulation to facilitate later fear extinction. Additional experiments ruled out stimulus familiarity as an explanation for the facilitative effects of IL stimulation and demonstrated that IL-mediated facilitations occur in procedures other than fear extinction. Together, these findings demonstrate that the IL stores inhibitory memories that are extremely flexible since they can be retrieved and used across many inhibitory procedures and distinct motivational contexts. These features establish the IL as a critical hub for the flexible application of inhibitory knowledge that allows adaptive responses in dynamic environments.

Introduction

Animals learn about environmental cues that predict the omission of important biological events (Sosa, 2024 [DOI](#)). The neural mechanisms underlying this inhibitory learning have been extensively investigated through studies of fear extinction, a process in which repeated exposure to a fear-inducing stimulus without its associated threat gradually diminishes the stimulus' capacity to elicit fear responses (Rescorla, 2000 [DOI](#); Delamater, 2004 [DOI](#)). These studies identified the infralimbic cortex (IL) in rodents as a critical neural hub for fear extinction learning (Bouton et al., 2021 [DOI](#); Quirk & Mueller, 2008 [DOI](#); Milad & Quirk, 2012 [DOI](#); Maren & Quirk, 2004 [DOI](#); Lingawi et al., 2019 [DOI](#);

Nett & LaLumiere, 2021 [↗](#); Marek et al., 2018b [↗](#)). Disrupting IL function across extinction training impairs extinction learning and increases fear responses during later testing (Do-Monte et al., 2015 [↗](#); Lebrón et al., 2004 [↗](#); Sierra-Mercado et al., 2011 [↗](#); Laurent & Westbrook, 2009 [↗](#)). Moreover, enhanced firing in IL neurons correlates with improved extinction retrieval (Milad & Quirk, 2002 [↗](#); Burgos-Robles et al., 2007 [↗](#)), while suppressing IL activity undermines this retrieval process (Laurent & Westbrook, 2009 [↗](#); Kim et al., 2016 [↗](#); Marek et al., 2018a [↗](#)). Conversely, artificially stimulating IL activity enhances extinction memory retrieval (Do-Monte et al., 2015 [↗](#); Kim et al., 2016 [↗](#); Milad et al., 2004 [↗](#); Vidal-Gonzalez et al., 2006 [↗](#)), providing compelling evidence for the IL's fundamental role in the fear inhibition learned in extinction. However, these findings leave critical questions about the specific nature of the memory being formed in extinction unanswered. Identifying the precise content of inhibitory learning (e.g., what exactly animals learn when they undergo fear extinction) remains essential for understanding both the mechanisms of inhibition and the broader adaptive functions of the IL in behavioral flexibility.

To study the content of inhibitory learning, we examined some of the conditions that allow retrieval of an inhibitory memory stored in the IL. We reasoned that successful retrieval could be revealed by demonstrating that IL stimulation during memory recall durably enhances the retrieved memory (Lingawi et al., 2018 [↗](#); Lingawi et al., 2017 [↗](#)). We found that pharmacological stimulation of the IL during an initial fear extinction session suppressed fear responses during that session but did not enhance fear inhibition during later testing. However, the same stimulation applied during a *second* extinction session significantly enhanced fear inhibition at test. These results demonstrated that prior fear extinction experience enabled IL stimulation to facilitate fear inhibition during subsequent extinction learning. The facilitation effect proved to be stimulus-specific, occurring only when the same fear-inducing stimulus was used across both extinction sessions. Additional experiments revealed that prior fear extinction was not the only experience that enabled IL stimulation to facilitate fear extinction. Similar facilitative effects emerged when animals were pre-exposed to the stimulus in a latent inhibition procedure or when the stimulus had previously predicted an appetitive outcome, such as food, followed by extinction of that prediction. These findings led us to propose that inhibitory procedures generate stimulus-specific inhibitory memories within the IL that lack motivational information about the omitted event. During subsequent fear extinction, IL stimulation can retrieve and strengthen these pre-existing inhibitory memories, thereby facilitating fear inhibition.

Our proposal aligns with research demonstrating IL involvement across diverse inhibitory learning procedures beyond those discussed above (Rhodes & Killcross, 2007 [↗](#); Fam et al., 2022 [↗](#); Ng et al., 2023 [↗](#); Krueger et al., 2023 [↗](#); Ng & Sangha, 2022 [↗](#); Broomer & Bouton, 2024 [↗](#)). The hypothesis that IL memories lack motivational specificity is consistent with evidence showing IL engagement in procedures involving non-threatening outcomes (Peters et al., 2009 [↗](#); Nett & LaLumiere, 2021 [↗](#); Laurent et al., 2016 [↗](#); Meyer & Bucci, 2014 [↗](#); Brown et al., 2023 [↗](#); Broomer & Bouton, 2024 [↗](#)), suggesting a general role in inhibitory processing rather than fear-specific mechanisms. Nevertheless, the findings described have at least three limitations that warrant consideration. First, all the first-phase inhibitory procedures involved presenting the target stimulus in isolation, raising questions about whether IL recruitment occurs under different stimulus presentation conditions. Some existing literature suggests this may indeed be a critical factor in IL engagement (Lay et al., 2020 [↗](#)). Second, prior experience with fear extinction, latent inhibition, or appetitive extinction could have simply increased familiarity with the target stimulus, and this familiarity might have enabled IL stimulation to be effective during subsequent fear extinction. Finally, the facilitative effects of IL stimulation were demonstrated exclusively during fear extinction sessions, limiting the generalizability of the findings to other inhibitory learning procedures.

The present series of experiments addressed these limitations to enhance understanding of inhibitory learning mechanisms and IL function. To determine whether IL engagement extends beyond procedures employing exposures to the target stimulus in isolation, the experiments utilized backward conditioning, during which a target stimulus is reliably presented a few seconds

after encountering a biologically significant event. This procedure has been shown to imbue the backward conditioned stimulus with inhibitory properties (Laurent et al., 2022 [DOI](#); Laurent et al., 2018 [DOI](#); Laurent et al., 2015 [DOI](#); Laurent & Balleine, 2015 [DOI](#); Delamater et al., 2003 [DOI](#); Seitz et al., 2022 [DOI](#); Sosa, 2024 [DOI](#); Chang et al., 2003 [DOI](#); Cole & Miller, 1999 [DOI](#); Heth, 1976 [DOI](#)). Thus, the first key experiment in the series examined whether prior experience with backward fear conditioning enables IL stimulation to facilitate subsequent fear extinction. After showing that this is, indeed, the case subsequent experiments then examined whether the facilitative effects of IL stimulation can be observed following backward *appetitive* conditioning, is due to familiarity of the stimulus that would eventually undergo extinction and can be observed across different circumstances of testing. Since the current research was interested in uncovering the general role played by the IL in inhibitory learning, it used an optogenetic approach that stimulated all local neurons regardless of their molecular identity. This broad approach more closely mimicked the pharmacological stimulation used in our previous studies (Lingawi et al., 2018 [DOI](#); Lingawi et al., 2017 [DOI](#)), while ensuring that the stimulation was restricted anatomically to the IL and temporally aligned with presentations of the target stimulus. The first two experiments therefore involved ex-vivo electrophysiological assessments of the capacity to stimulate IL neurons, followed by a replication of the core finding that prior fear extinction experience enables IL stimulation to enhance subsequent fear extinction learning.

Results

Ex-vivo cell recordings demonstrate successful optical IL stimulation

We first used ex-vivo electrophysiological recordings to assess our capacity to optically stimulate IL neurons. Wild-type Long-Evans female and male rats were bilaterally infused in the IL with an excitatory channelrhodopsin (**ChR2**) virus (Figure 1A [DOI](#)), and recordings were completed in 10 cells transduced with the virus. Overall, LED light illumination (465 nm, 5 ms pulse, 20 Hz) enhanced action potential firing compared to the period preceding illumination (baseline; Figure 1B [DOI](#); $F_{(1,9)} = 13.69$, $p < 0.01$; $\eta^2 = 0.60$). Based on the physiological and morphological characteristics of these cells, we identified one as a non-pyramidal cell, which was excited by LED light illumination (from 0 at baseline to 13.6 Hz under LED). The remaining 9 cells were pyramidal, and one did not respond to LED light illumination, but we cannot exclude the possibility that this was due to a lack of ChR2 expression in the somatic compartment. Another pyramidal cell showed a slight reduction in activity following LED light illumination (from 6 Hz at baseline to 3.6 Hz under LED), but the remaining 7 cells displayed clear excitation upon LED light illumination. Thus, we conclude that ChR2-mediated excitation was substantial at the time of LED light delivery, effectively validating our ability to use ChR2 to stimulate IL neuronal activity.

Prior experience with fear extinction enables IL stimulation to facilitate subsequent fear extinction

Experiment 1 aimed to confirm that IL stimulation facilitates fear extinction in animals with previous extinction experience (Lingawi et al., 2017 [DOI](#)). Rats received bilateral IL infusions of the ChR2 virus and were implanted with fiber-optic cannulae positioned above the IL (Figure 2-figure supplement 1A [DOI](#)). All animals then underwent the behavioral procedures described in Figure 2A [DOI](#). Following initial forward fear conditioning to a tone (i.e., the tone ended with shock delivery), rats were allocated to two conditions: half (ReExt-OFF and ReExt-ON) received an initial fear extinction session to the tone, while in the other condition (Ext-OFF and Ext-ON) rats were handled only. All rats subsequently underwent a second fear conditioning session followed by a second fear extinction session. In line with our previous work (Lingawi et al., 2018 [DOI](#); Lingawi et al., 2017 [DOI](#)), the second extinction session in this experiment and the subsequent ones was brief, involving 8 tone alone presentations (6 presentations in remaining experiments). The low number of stimulus presentations aimed to generate minimal extinction learning to maximize detection of

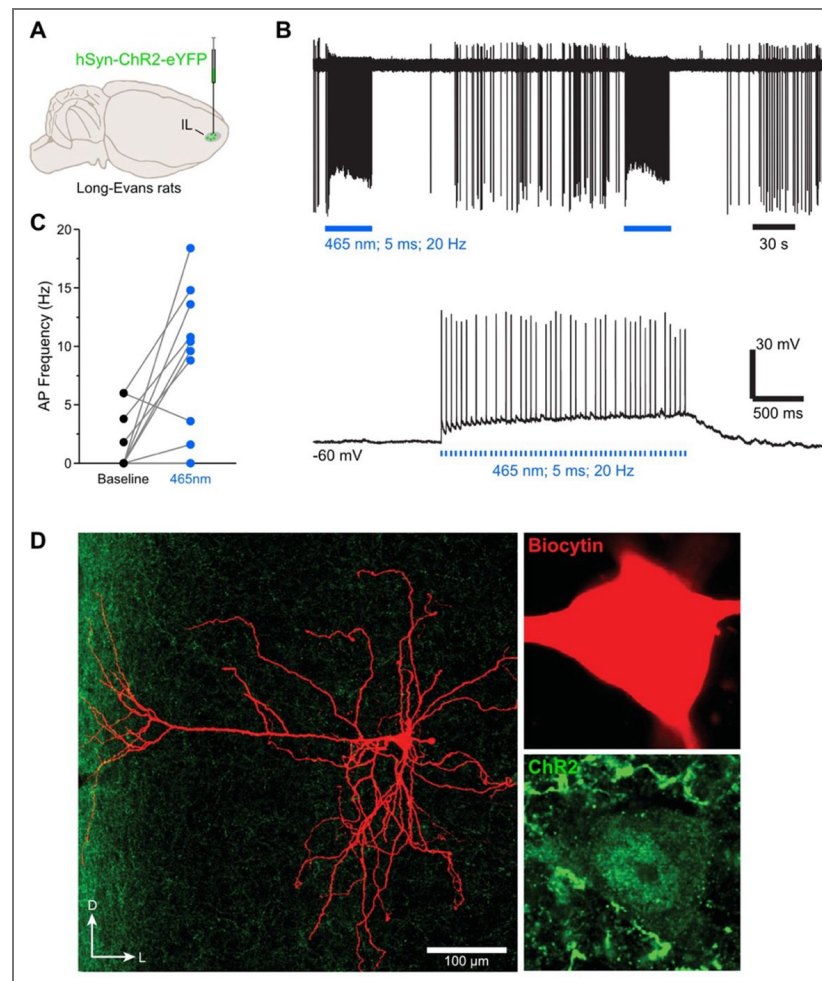


Figure 1. Ex-vivo cell recordings demonstrate successful optical IL stimulation

A. Wild-type Long-Evans rats ($n = 2$ females and 2 males) were bilaterally infused in the IL with a ChR2 virus. **B.** The top representative raw trace is a cell-attached recording from a transfected IL neuron with LED train stimulations. In the same neuron under whole-cell current clamp, LED pulses excited the IL neuron as evidenced by enhanced action potentials (AP) from resting state baseline (bottom raw trace). **C.** AP frequency was enhanced during LED illumination compared to their baseline state ($n = 10$ cells). **D.** Micrographs showing a ChR2 transfected IL neurons that was labelled with biocytin.

any facilitatory effects of IL stimulation. During the brief extinction session, LEDs were activated in half the animals from each behavioral condition (Ext-ON and ReExt-ON) but remained inactive in the other half (Ext-OFF and ReExt-OFF). Finally, all rats were tested for freezing to the tone.

Freezing data are presented in Figure 2B. Initial forward fear conditioning proceeded as expected, with freezing to the tone gradually increasing across trials (Trial: $F_{(1,34)} = 195.08$; $p < 0.001$; $\eta^2 = 0.85$). A marginal protocol effect emerged, with rats designated for two extinction sessions (ReExt-OFF and ReExt-ON) showing greater freezing than those assigned to a single session (Ext-OFF and Ext-ON; Protocol: $F_{(1,34)} = 4.56$; $p < 0.05$; $\eta^2 = 0.12$). Nevertheless, all rats increased their freezing to the tone as training progressed, regardless of protocol assignment (Protocol x Trial: $p = 0.09$). No other significant differences were observed (lowest $p = 0.45$), including during the baseline period (i.e., prior to the first tone presentation; lowest $p = 0.13$).

The first extinction session proceeded smoothly, with tone-elicited freezing gradually decreasing across trial blocks (Block: $F_{(1,17)} = 56.225$; $p < 0.05$; $\eta^2 = 0.77$) regardless of subsequent LED condition assignment (Block x LED: $p = 0.49$). No other significant differences were detected (LED: $p = 0.07$), including during the baseline period (LED: $p = 0.36$).

The second forward conditioning session was successful, with freezing to the tone increasing across trials (Trial: $F_{(1,34)} = 11.41$; $p < 0.01$; $\eta^2 = 0.25$) regardless of protocol or LED assignment (lowest $p = 0.09$). No other significant differences were detected (lowest $p = 0.47$), including during the baseline period (lowest $p = 0.26$). It is worth noting that these analyses indicate that the marginal effect of protocol observed during initial forward fear conditioning or the trend for lower freezing in the ReExt-ON condition during the first extinction session were short-lived.

The brief extinction session produced a gradual decline in tone-elicited freezing across trials (Trial: $F_{(1,34)} = 12.32$; $p < 0.001$; $\eta^2 = 0.25$) regardless of protocol assignment (Protocol x Trial: $p = 0.88$). IL stimulation severely reduced freezing (LED: $F_{(1,34)} = 67.82$; $p < 0.001$; $\eta^2 = 0.67$) regardless of protocol assignment (LED x Protocol: $p = 0.39$). Consequently, the gradual decline in freezing across trial blocks was more pronounced in rats that did not receive IL stimulation (LED x Trial: $F_{(1,34)} = 4.81$; $p < 0.05$; $\eta^2 = 0.12$). No other significant differences were observed (lowest $p = 0.66$), including during the baseline period (lowest $p = 0.09$).

The test data confirmed that IL stimulation during the brief extinction session reduced freezing only in rats that had previously undergone an extinction session (Figure 2B and Figure 2-figure supplement 1B). Rats without prior extinction experience displayed substantial and equivalent freezing levels regardless of IL stimulation (Ext-OFF vs. Ext-ON: $p = 0.96$). These freezing levels were comparable to those observed in rats that had received initial extinction but no IL stimulation during the brief session (Ext-OFF + Ext-ON vs. ReExt-OFF: $p = 0.25$). Critically, all three of these groups exhibited significantly greater freezing than rats that had received both initial extinction and IL stimulation during the subsequent brief extinction session (Ext-OFF + Ext-ON + ReExt-OFF vs. ReExt-ON: $F_{(1,34)} = 9.20$; $p < 0.01$; $\eta^2 = 0.21$). No significant differences emerged during the baseline period (lowest $p = 0.21$).

These findings confirm that prior fear extinction experience enables IL stimulation to enhance fear extinction (Lingawi et al., 2017). This pattern aligns with the proposal that the IL encodes inhibitory memories formed during initial extinction training, and that IL stimulation during later sessions promotes the retrieval and strengthening of these original inhibitory memories.

Prior experience with backward fear conditioning enables IL stimulation to facilitate subsequent fear extinction

Experiment 2 tested whether prior experience with backward fear conditioning enables IL stimulation to facilitate fear extinction. Since Experiment 1 showed no effect of LED activation alone, we eliminated this factor and instead included a null virus control. Rats received bilateral IL infusions of a null (EYFP) or Chr2 virus and were implanted with fiber-optic cannulae positioned above the IL (Figure 3-figure supplement 1A). As outlined in Figure 3A, half the rats in each viral condition (Back-EYFP and Back-Chr2) underwent backward fear conditioning to a tone (i.e., the tone was presented 5 seconds after shock delivery), while the remaining rats (No-

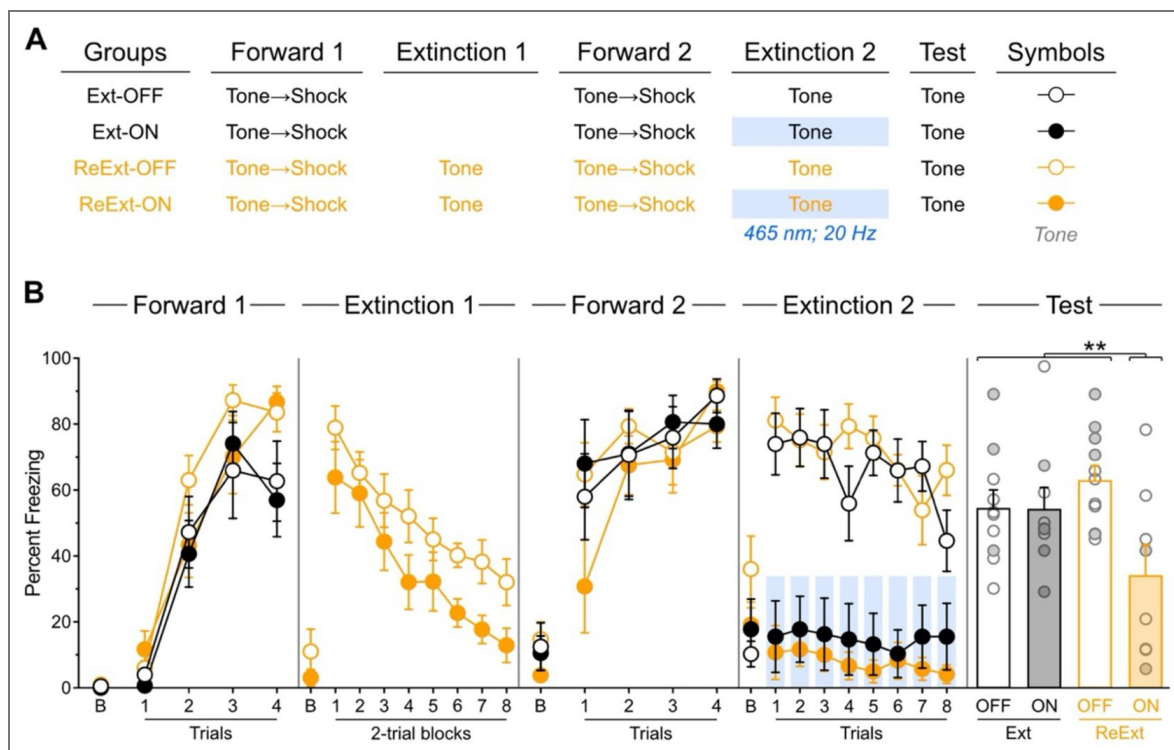


Figure 2. Prior experience with fear extinction enables IL stimulation to facilitate subsequent fear extinction

A. Schematic representation of the behavioral design used in Experiment 1. The LEDs were activated in the Ext-ON and ReExt-ON groups during the second fear extinction session. Ext-OFF: $n = 3$ females and 7 males; Ext-ON: $n = 4$ females and 5 males; ReExt-OFF: $n = 5$ females and 6 males; ReExt-ON: $n = 2$ females and 6 males. **B.** Mean percent freezing during all experimental stages. Data are shown as mean $\pm$ SEM. Test data include individual data points for female (filled circle) and male (open circle) rats. Asterisks denote significant effect (** $p < 0.01$).

EYFP and No-ChR2) received handling only. Previous research indicates that inhibition in backward conditioning is more likely to develop with prolonged training and with the insertion of a time delay between delivery of the motivational event (e.g., the footshock) and presentation of the stimulus (Laurent et al., 2022 [DOI](#); Laurent et al., 2018 [DOI](#); Laurent et al., 2015 [DOI](#); Laurent & Balleine, 2015 [DOI](#); Delamater et al., 2003 [DOI](#); Seitz et al., 2022 [DOI](#); Sosa, 2024 [DOI](#); Chang et al., 2003 [DOI](#); Cole & Miller, 1999 [DOI](#); Heth, 1976 [DOI](#)). Thus, backward conditioning in this experiment and the next experiments was administered across five consecutive days and using a 5-second delay between shock presentation and tone onset. After this conditioning, all rats underwent forward tone fear conditioning followed by a brief fear extinction session with LED activation during all tone presentations. Finally, all rats were tested for freezing to the tone.

Freezing data are presented in Figure 3B [DOI](#). Backward fear conditioning proceeded as expected (Laurent et al., 2022 [DOI](#)), with overall freezing increasing across days (Day: $F_{(1,17)} = 27.54$; $p < 0.001$; $\eta^2 = 0.62$). Rats showed greater freezing during the pre-shock periods compared to tone presentations (Period: $F_{(1,17)} = 76.09$; $p < 0.001$; $\eta^2 = 0.82$) with this discrimination becoming larger as training progressed (Period $\times$ Day: $F_{(1,17)} = 174.68$; $p < 0.001$; $\eta^2 = 0.91$). Virus assignment had no significant influence on these effects (lowest $p = 0.13$).

Forward fear conditioning was successful, with freezing to the tone gradually increasing across trials (Trial: $F_{(1,32)} = 99.44$; $p < 0.001$; $\eta^2 = 0.76$). However, this increase was less pronounced in rats that had received backward conditioning (Trial $\times$ Protocol: $F_{(1,32)} = 14.72$; $p < 0.001$; $\eta^2 = 0.32$), as these animals displayed elevated freezing during the first tone trial. This pattern suggests that backward conditioning generated modest excitatory associations between the tone and shock (Cole & Miller, 1999 [DOI](#)). Evidence for backward conditioning effects was further supported by the baseline period analysis, where rats with backward conditioning experience showed significantly greater context freezing (Protocol: $F_{(1,32)} = 11.66$; $p < 0.01$; $\eta^2 = 0.27$). This indicates that backward conditioning established excitatory associations between the context (i.e., conditioning chamber) and shock (Chang et al., 2003 [DOI](#)). No other significant differences were observed during this stage (lowest $p = 0.45$).

The brief extinction session produced a gradual decline in freezing to the tone across trials (Trial: $F_{(1,32)} = 17.16$; $p < 0.001$; $\eta^2 = 0.55$) regardless of protocol assignment (Protocol $\times$ Trial: $p = 0.63$). Consistent with Experiment 1, IL stimulation reduced tone freezing (Virus: $F_{(1,32)} = 14.02$; $p < 0.001$; $\eta^2 = 0.30$) regardless of protocol assignment (Virus $\times$ Protocol: $p = 0.11$). However, unlike the previous experiment, the gradual decline in freezing across tone trials was equivalent between IL-stimulated and non-stimulated rats (Virus $\times$ Trial: $p = 0.07$). Visual inspection of the data suggests this lack of difference resulted from reduced freezing in non-IL-stimulated rats that had received backward fear conditioning. This reduction was not unexpected, given that inhibitory stimuli acquire excitatory properties more slowly than neutral stimuli (Rescorla, 1969 [DOI](#)), and we anticipated that backward fear conditioning would confer inhibitory properties to the tone. A planned *a priori* analysis confirmed that non-IL-stimulated rats showed reduced freezing when they had received backward conditioning (Back-EYFP vs. No-EYFP: $F_{(1,32)} = 13.58$; $p < 0.001$; $\eta^2 = 0.30$). Consistent with the previous stage, baseline period analysis revealed greater context freezing in backward-conditioned rats (Protocol: $F_{(1,32)} = 19.82$; $p < 0.001$; $\eta^2 = 0.38$), confirming substantial excitatory associations between the context and shock. No other significant differences were observed during this stage (lowest $p = 0.11$).

The test data revealed that IL stimulation during the brief fear extinction session reduced freezing only in rats that had undergone backward conditioning (Figure 3B [DOI](#) and Figure 3-figure supplement 1B [DOI](#)). Rats without prior backward experience displayed substantial and equivalent freezing levels regardless of IL stimulation (No-EYFP vs. No-ChR2: $p = 0.28$). These freezing levels were comparable to those observed in rats that had received backward conditioning but no IL stimulation during the brief extinction session (No-EYFP + No-ChR2 vs. Back-EYFP: $p = 0.70$). Critically, all three of these groups exhibited significantly greater freezing than rats that had received both backward conditioning and IL stimulation during the brief extinction session (No-EYFP + No-ChR2 + Back-EYFP vs. Back-ChR2: $F_{(1,32)} = 8.15$; $p < 0.01$; $\eta^2 = 0.21$). The baseline period

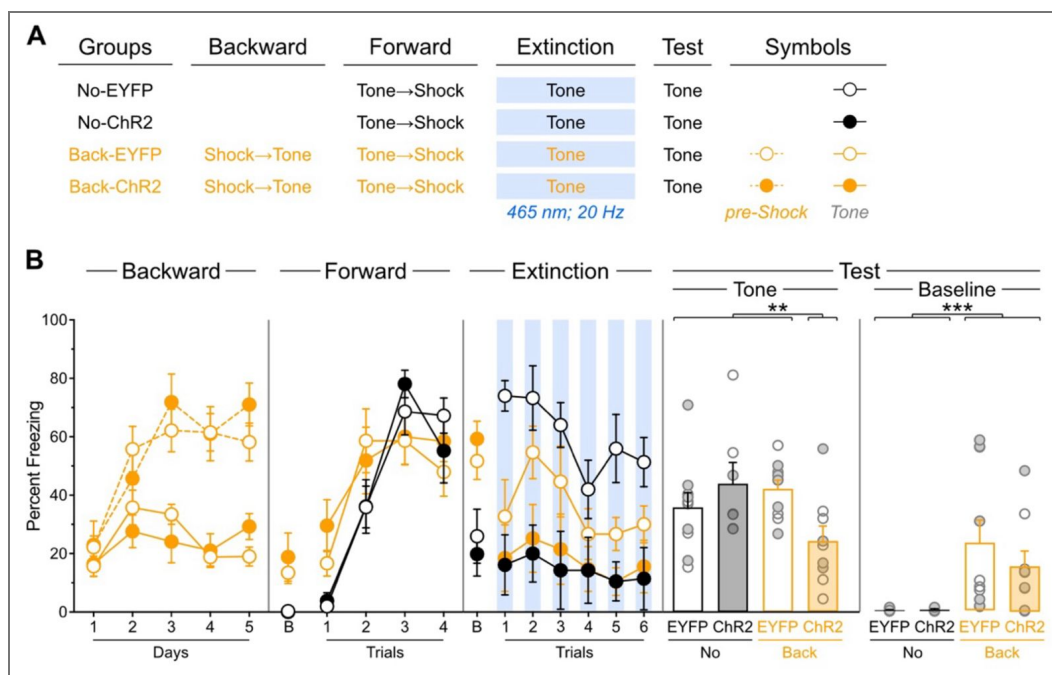


Figure 3. Prior experience with backward fear conditioning enables IL stimulation to facilitate subsequent fear extinction

A. Schematic representation of the behavioral design used in Experiment 2. The LEDs were activated in all groups during the fear extinction session. No-EYFP: $n = 5$ females and 5 males; No-ChR2: $n = 3$ females and 4 males; Back-EYFP: $n = 5$ females and 5 males; Back-ChR2: $n = 4$ females and 5 males. **B.** Mean percent freezing during all experimental stages. Baseline freezing (before first tone presentation) at test is provided due to significant differences. Data are shown as mean $\pm$ SEM. Test data include individual data points for female (filled circle) and male (open circle) rats. Asterisks denote significant effect (** $p < 0.01$; *** $p < 0.001$).

analysis revealed greater context freezing in rats that received backward conditioning (Protocol: $F_{(1,32)} = 13.49$; $p < 0.001$; $\eta^2 = 0.30$) irrespective of virus assignment (Protocol x Virus: $p = 0.46$). Virus assignment had no effect during the baseline period ($p = 0.46$).

These findings indicate that backward fear conditioning enables IL stimulation to facilitate subsequent fear extinction. Thus, presenting the target stimulus in isolation does not appear to be the critical factor controlling IL engagement in memory encoding. Rather, these findings are consistent with the hypothesis that backward fear conditioning establishes inhibitory memories in the IL that can be retrieved and strengthened during subsequent fear extinction sessions through local optogenetic stimulation.

Backward fear conditioning produces an IL memory that is specific to the backward trained stimulus and is inhibitory

Experiment 3 addressed two key questions. First, it tested whether the facilitative effect of IL stimulation following backward fear conditioning operates through modulation of baseline contextual freezing, since differences in such freezing were present in the previous experiment. Second, it examined the stimulus specificity of the IL-mediated facilitation by determining whether the effect is restricted to the stimulus that underwent backward fear conditioning. All rats received bilateral IL infusions of either EYFP or ChR2 virus and were implanted with fiber-optic cannulae targeting the IL (Figure 4-figure supplement 1A). Following the procedures shown in Figure 4A, rats in each viral group were randomly assigned to one of two backward conditioning procedures: half received backward conditioning with a light (Diff-EYFP and Diff-ChR2 groups), while the other half received identical conditioning using a tone (Same-EYFP and Same-ChR2 groups). Subsequently, all rats underwent standard forward tone fear conditioning, followed by a brief extinction session during which LED stimulation was delivered concurrent with each tone presentation. Finally, all rats were tested for freezing to the tone.

Freezing data are presented in Figure 4B. Backward fear conditioning proceeded as expected, with overall freezing increasing across days (Day: $F_{(1,22)} = 35.02$; $p < 0.001$; $\eta^2 = 0.61$). Rats showed greater freezing during the pre-shock periods compared to stimulus presentations (Period: $F_{(1,22)} = 139.81$; $p < 0.001$; $\eta^2 = 0.86$) with this discrimination becoming larger as training progressed (Period x Day: $F_{(1,22)} = 98.77$; $p < 0.001$; $\eta^2 = 0.82$). An unexpected Protocol x Virus interaction emerged ($F_{(1,22)} = 4.43$; $p < 0.05$; $\eta^2 = 0.17$). In the rats trained with the light (Diff-EYFP and Diff-ChR2 groups), overall freezing was similar across virus condition (Virus: $p = 0.50$). It increased across days (Day: $F_{(1,11)} = 22.54$; $p < 0.001$; $\eta^2 = 0.67$), it was greater in the pre-shock periods compared to stimulus presentation (Period: $F_{(1,11)} = 80.53$; $p < 0.001$; $\eta^2 = 0.88$), and the discrimination between the two periods became larger as training progressed (Day x Period: $F_{(1,11)} = 59.46$; $p < 0.001$; $\eta^2 = 0.84$). None of these effects were influenced by viral assignment (Virus x Day, Virus x Period, Virus x Day x Period: lowest $p = 0.28$). In the rats trained with the tone (Same-EYFP and Same-ChR2 groups), overall freezing was greater in those that had been infused with the ChR2 virus (Virus: $F_{(1,11)} = 9.49$; $p < 0.05$; $\eta^2 = 0.46$). Freezing increased across days (Day: $F_{(1,11)} = 13.50$; $p < 0.001$; $\eta^2 = 0.55$), it was higher in the pre-shock periods compared to stimulus presentation (Period: $F_{(1,11)} = 66.33$; $p < 0.001$; $\eta^2 = 0.86$) and the discrimination between the two periods grew as training progressed (Day x Period: $F_{(1,11)} = 41.02$; $p < 0.001$; $\eta^2 = 0.79$). None of these factors were influenced by virus assignment (Virus x Day, Virus x Period, Virus x Day x Period: lowest $p = 0.10$). No other significant differences were observed during this stage (lowest $p = 0.06$).

Forward fear conditioning was successful, with freezing to the tone gradually increasing across trials (Trial: $F_{(1,22)} = 58.104$; $p < 0.001$; $\eta^2 = 0.73$). A Protocol x Virus interaction was again detected ($F_{(1,22)} = 9.28$; $p < 0.01$; $\eta^2 = 0.30$). In the rats trained with the light (Diff-EYFP and Diff-ChR2 groups), overall freezing was similar across virus condition (Virus: $p = 0.22$) and increased across trials (Trial: $F_{(1,11)} = 64.18$; $p < 0.001$; $\eta^2 = 0.85$) regardless of viral assignment (Virus x Trial: $p = 0.52$). In the rats trained with the tone (Same-EYFP and Same-ChR2 groups), overall freezing was greater in rats that had been infused with ChR2 (Virus: $F_{(1,11)} = 11.63$; $p < 0.01$; $\eta^2 = 0.51$). Nevertheless, freezing increased across trials (Trial: $F_{(1,11)} = 12.45$; $p < 0.01$; $\eta^2 = 0.53$) regardless of

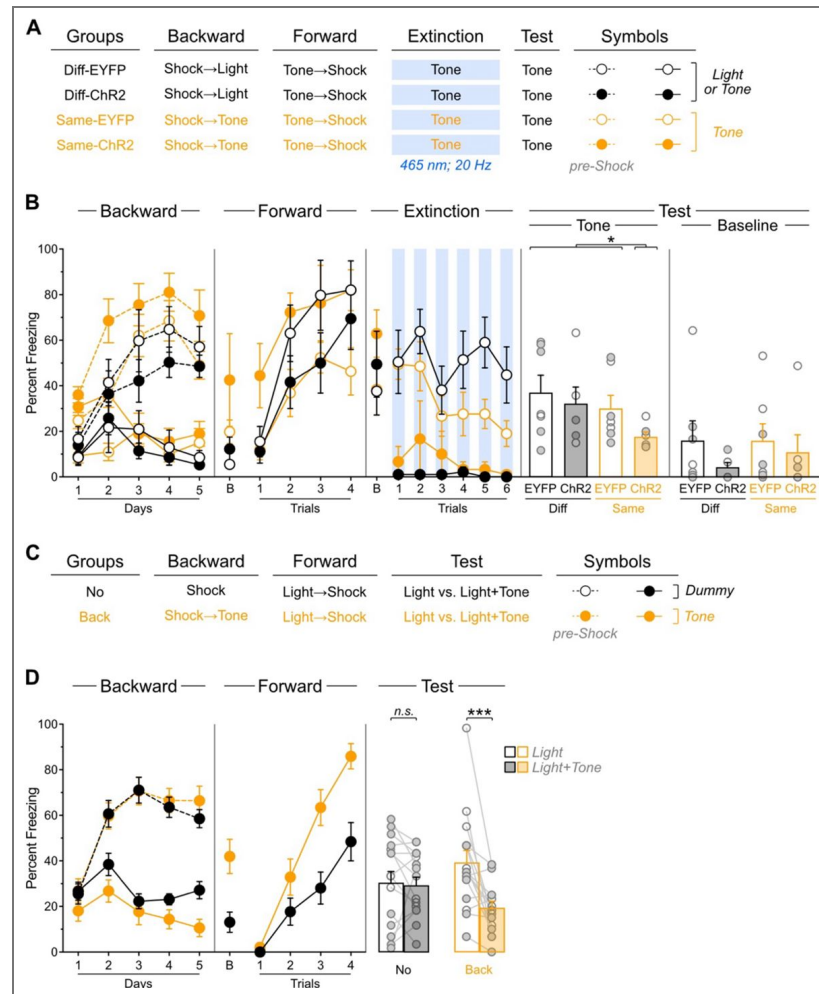


Figure 4. Backward fear conditioning produces an IL memory that is specific to the backward trained stimulus and is inhibitory

A. Schematic representation of the behavioral design used in Experiment 3. The LEDs were activated in all groups during the fear extinction session. Diff-EYFP: $n = 4$ females and 3 males; Diff-ChR2: $n = 2$ females and 4 males; Same-EYFP: $n = 4$ females and 3 males; Same-ChR2: $n = 2$ females and 4 males. **B.** Mean percent freezing during all experimental stages. Baseline freezing (before first tone presentation) at test is provided to compare with Experiment 3. **C.** Schematic representation of the behavioral design used in Experiment 4. No: $n = 12$ females and 4 males; Back: $n = 8$ females and 8 males. **D.** Mean percent freezing during all experimental stages. Data are shown as mean $\pm$ SEM. Test data include individual data points for female (filled circle) and male (open circle) rats. Asterisks denote significant effect (* $p < 0.05$; *** $p < 0.001$; n.s., nonsignificant).

viral assignment (Virus $\times$ Trial: $p = 0.97$). No other significant differences were observed during this stage (lowest $p = 0.06$). During the baseline period, rats that had previously received backward conditioning with the tone displayed greater freezing (Protocol: $F_{(1,22)} = 4.84$; $p < 0.05$; $\eta^2 = 0.18$). No other significant differences were observed during this period (lowest $p = 0.16$). Although some stimulus-related differences appeared, their interpretation remains unclear given their presence in the earlier experimental stage and the expectation that baseline contextual freezing should be independent of stimulus identity.

The brief extinction session produced a gradual decline in tone-elicited freezing across trials (Trial: $F_{(1,22)} = 6.33$; $p < 0.05$; $\eta^2 = 0.22$) regardless of protocol and viral assignment (Protocol $\times$ Trial: $p = 0.06$; Virus $\times$ Trial: $p = 0.23$), and the three factors did not interact (Protocol $\times$ Virus $\times$ Trial: $p = 0.37$). Consistent with previous experiments, IL stimulation reduced tone freezing (Virus: $F_{(1,22)} = 38.55$; $p < 0.001$; $\eta^2 = 0.64$) regardless of protocol assignment (Virus $\times$ Protocol: $p = 0.07$). In contrast to the previous experiment, no significant differences were detected during the baseline period (lowest $p = 0.13$).

The test results revealed that backward fear conditioning enables IL stimulation to facilitate subsequent fear extinction only when the same stimulus is employed throughout the procedures (Figure 4B [↗](#) and Figure 4-figure supplement 1B [↗](#)). Rats with prior backward experience to the light displayed substantial and equivalent freezing levels to the tone regardless of IL stimulation (Diff-EYFP vs. Diff-ChR2: $p = 0.28$). These freezing levels were comparable to those observed in rats that had received backward conditioning to the tone but no IL stimulation during the brief extinction session (Diff-EYFP + Diff-ChR2 vs. Same-EYFP: $p = 0.54$). Critically, all three of these groups exhibited significantly greater freezing than rats that had received both backward conditioning to the tone and IL stimulation during the brief extinction session (Diff-EYFP + Diff-ChR2 + Same-EYFP vs. Same-ChR2: $F_{(1,22)} = 4.41$; $p < 0.05$; $\eta^2 = 0.17$). Unlike in the previous experiment, no significant differences were observed during the baseline period (lowest $p = 0.26$). It is worth noting, however, that some unexpected interactions emerged across the first two training stages, which were driven by greater freezing in rats that had received backward conditioning to the tone and ChR2 infusion in the IL. Although the source of this greater freezing remains unclear, it seems unlikely to have influenced test performance, since the group concerned expressed lower freezing during that test.

These findings confirm that backward fear conditioning enables IL stimulation to facilitate subsequent fear extinction. Crucially, they also demonstrate that this facilitative effect operates independently of baseline contextual freezing modulation: rather it is specific to the stimulus that underwent backward conditioning. Thus, backward fear conditioning generates stimulus-specific memories within the IL.

The interpretation of the previous results depends on the hypothesis that backward fear conditioning establishes inhibitory memories that are subsequently retrieved during fear extinction. The ability of backward fear conditioning to confer inhibitory properties to the trained stimulus is supported by results from Experiment 2, where the backward-trained tone showed delayed acquisition of excitatory properties during forward fear conditioning, as evidenced by reduced freezing during brief extinction. This retardation effect was replicated in Experiment 3: the backward fear conditioned tone elicited less freezing during the brief fear extinction session compared to the neutral tone (Same-EYFP vs. Diff-EYFP: $F_{(1,22)} = 4.65$; $p < 0.05$; $\eta^2 = 0.17$). However, demonstrating inhibition requires additional evidence beyond retardation (Rescorla, 1969 [↗](#)), and Experiment 4 aimed to provide this by examining whether our backward protocol enables the stimulus to produce subtractive summation—the capacity of an inhibitory stimulus to reduce conditioned responding elicited by an excitatory stimulus. As illustrated in Figure 4C [↗](#), rats were randomly assigned to one of two procedures: rats in one procedure received backward fear conditioning to a tone (Back), while those in the other procedure received context shock exposure without the tone (No). Subsequently, all rats underwent forward fear conditioning to a light, followed by testing the next day. The test compared freezing responses to the light alone versus combined light and tone presentations.

Freezing data are presented in Figure 4D [↗](#). During the initial experimental phase, we monitored freezing during pre-shock periods for both groups, during tone presentations in the backward conditioning group, and during equivalent dummy periods in the control group (No). Overall freezing increased progressively across training days (Day: $F_{(1,30)} = 14.10$; $p < 0.001$; $\eta^2 = 0.32$) regardless of protocol assignment (Protocol $\times$ Day: $p = 0.192$). Rats showed greater freezing during pre-shock periods compared to tone presentations or dummy periods (Period: $F_{(1,30)} = 197.471$; $p < 0.001$; $\eta^2 = 0.87$), but this difference was more pronounced in rats that had received backward training (Protocol $\times$ Period: $F_{(1,30)} = 6.14$; $p < 0.05$; $\eta^2 = 0.17$). The discrimination between periods became larger as training progressed (Period $\times$ Day: $F_{(1,30)} = 82.20$; $p < 0.001$; $\eta^2 = 0.73$) regardless of protocol (Protocol $\times$ Period $\times$ Day: $p = 0.19$). Overall freezing levels were comparable across groups (Protocol: $p = 0.40$).

Forward fear conditioning to the light proceeded successfully, with freezing increasing gradually across trials (Trial: $F_{(1,30)} = 177.68$; $p < 0.001$; $\eta^2 = 0.86$). The increase was more pronounced in rats with prior backward fear conditioning (Protocol $\times$ Trial: $F_{(1,30)} = 14.85$; $p < 0.001$; $\eta^2 = 0.83$), suggesting that excitatory associations between the tone and shock from backward conditioning somehow enhanced fear to the light. Consequently, overall freezing was elevated in rats that had received backward fear conditioning (Protocol: $F_{(1,30)} = 11.71$; $p < 0.001$; $\eta^2 = 0.28$). During the baseline period, a similar pattern emerged, with backward-trained rats displaying greater contextual freezing (Protocol: $F_{(1,30)} = 10.80$; $p < 0.001$; $\eta^2 = 0.26$).

The test results revealed that backward fear conditioning produced subtractive summation. Although overall freezing was greater during light-alone presentations than during combined light and tone presentations (Period: $F_{(1,30)} = 11.32$; $p < 0.001$; $\eta^2 = 0.27$), this difference depended on protocol assignment (Protocol $\times$ Period: $F_{(1,30)} = 9.14$; $p < 0.01$; $\eta^2 = 0.23$). Simple effects analyses showed that control rats displayed equivalent freezing during both test periods ($p = 0.81$). In contrast, rats with prior backward fear conditioning experience showed significantly more freezing during light-alone presentations than during combined light and tone presentations ($F_{(1,30)} = 20.41$; $p < 0.001$; $\eta^2 = 0.40$). No other significant differences were observed (lowest $p = 0.93$), including during the baseline period (lowest $p = 0.66$).

This experiment demonstrates that backward fear conditioning endowed the tone stimulus with inhibitory properties capable of reducing freezing to the excitatory light stimulus. This result is consistent with the hypothesis that backward fear conditioning protocol establishes inhibitory memories within the IL that can be retrieved and strengthened during subsequent fear extinction sessions by optogenetic stimulation of the IL during subsequent fear extinction.

Inhibitory memories encoded within the IL lack motivational information

Experiment 5 tested whether IL memories incorporate detailed information about the biologically significant event used in backward conditioning. Rats received the same surgical procedures (Figure 5-figure supplement 1A [↗](#)) as those from Experiment 3, and identical behavioral procedures with one critical modification (Figure 5A [↗](#)): during the initial experimental stage, rats received backward appetitive conditioning (the food outcome was delivered and 5 seconds after entries in the magazine where it was delivered, the stimulus was turned on). Hence, rats in each viral group were randomly assigned to one of two backward appetitive conditioning procedures: in one procedure rats received backward conditioning with a light (Diff-EYFP and Diff-ChR2 groups), while those in the other procedure received identical conditioning using a tone (Same-EYFP and Same-ChR2 groups).

Appetitive performance is displayed as the rate of entries in the magazine where food was delivered (Figure 5B [↗](#)). Overall entries increased across training days (Day: $F_{(1,31)} = 5.91$; $p < 0.05$; $\eta^2 = 0.16$). Animals showed greater entries during stimulus presentations compared to pre-food periods (Period: $F_{(1,31)} = 79.64$; $p < 0.001$; $\eta^2 = 0.72$), with this discrimination disappearing as training progressed (Period $\times$ Day: $F_{(1,31)} = 20.31$; $p < 0.001$; $\eta^2 = 0.40$). An unexpected Protocol $\times$ Period interaction emerged ($F_{(1,31)} = 4.37$; $p < 0.05$; $\eta^2 = 0.12$). In the rats trained with the light (Diff-

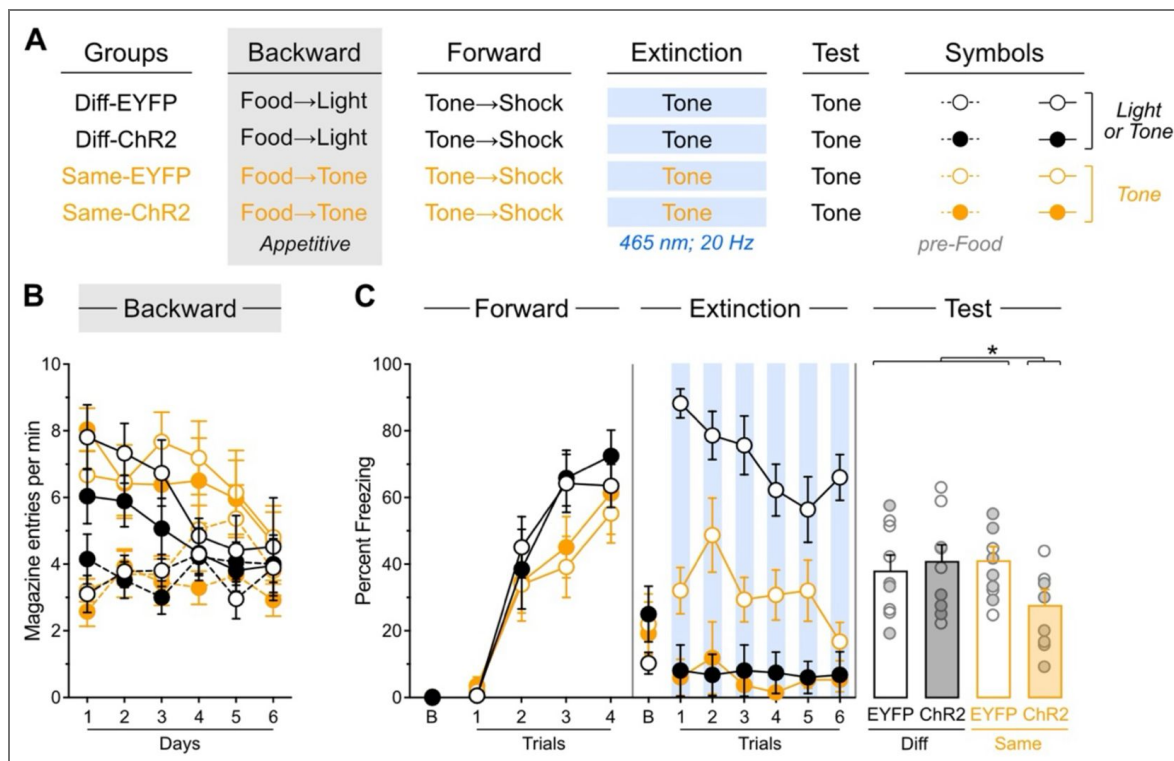


Figure 5. Inhibitory memories encoded within the IL lack motivational information

A. Schematic representation of the behavioral design used in Experiment 5. The LEDs were activated in all groups during the fear extinction session. Diff-EYFP: $n = 4$ females and 5 males; Diff-ChR2: $n = 4$ females and 5 males; Same-EYFP: $n = 6$ females and 3 males; Same-ChR2: $n = 6$ females and 2 males. **B.** Number of magazine entries during backward appetitive conditioning. **C.** Mean percent freezing during remaining experimental stages. Data are shown as mean $\pm$ SEM. Test data include individual data points for female (filled circle) and male (open circle) rats. Asterisks denote significant effect ($* p < 0.05$).

EYFP and Diff-ChR2 groups), overall entries were similar across virus condition (Virus: $p = 0.41$). They increased across days (Day: $F_{(1,16)} = 5.97$; $p < 0.05$; $\eta^2 = 0.27$), were greater during stimulus presentations compared to pre-food periods (Period: $F_{(1,16)} = 24.02$; $p < 0.001$; $\eta^2 = 0.76$), with this discrimination disappearing as training progressed (Period $\times$ Day: $F_{(1,16)} = 10.46$; $p < 0.01$; $\eta^2 = 0.40$). None of these factors were influenced by viral assignment (Virus $\times$ Day, Virus $\times$ Period, Virus $\times$ Day $\times$ Period: lowest $p = 0.08$). In the rats trained with the tone (Same-EYFP and Same-ChR2 groups), overall entries were similar across virus condition (Virus: $p = 0.06$). They remained stable across days (Day: $p = 0.25$), were greater during stimulus presentations compared to pre-food periods (Period: $F_{(1,15)} = 6.28$; $p < 0.05$; $\eta^2 = 0.30$), with this discrimination disappearing as training progressed (Period $\times$ Day: $F_{(1,15)} = 152.18$; $p < 0.001$; $\eta^2 = 0.91$). None of these factors were influenced by viral assignment (Virus $\times$ Day, Virus $\times$ Period, Virus $\times$ Day $\times$ Period: lowest $p = 0.13$). No other significant differences were observed during this stage (lowest $p = 0.06$). It is noteworthy that the pattern of stimulus-elicited response during backward appetitive responding differ substantially from that observed during backward fear conditioning. However, the appetitive performance observed is like that obtained in previous studies (Laurent et al., 2017 [DOI](#); Seitz et al., 2022 [DOI](#); Laurent et al., 2018 [DOI](#); Laurent & Balleine, 2015 [DOI](#)) and should not be taken as a failure of the protocol to generate inhibitory learning. It also suggests that the unexpected interactions in this and previous stages are unlikely to have uncovered any influence on that learning.

Freezing data are presented in Figure 5C [DOI](#). Forward fear conditioning proceeded successfully, with tone-elicited freezing increasing gradually across trials (Trial: $F_{(1,31)} = 240.95$; $p < 0.001$; $\eta^2 = 0.89$). No other significant differences were observed (lowest $p = 0.06$), including during the baseline period (lowest $p = 0.26$).

The brief fear extinction session produced a gradual decline in tone-elicited freezing across trials (Trial: $F_{(1,31)} = 10.43$; $p < 0.01$; $\eta^2 = 0.25$) regardless of protocol assignment (Protocol: $p = 0.49$). Consistent with previous experiments, IL stimulation significantly reduced tone freezing (Virus: $F_{(1,31)} = 178.70$; $p < 0.001$; $\eta^2 = 0.85$), and consequently, the gradual decline in freezing across trials was more pronounced in non-stimulated rats (Virus $\times$ Trial: $F_{(1,31)} = 12.52$; $p < 0.001$; $\eta^2 = 0.29$). Furthermore, the freezing reduction produced by IL stimulation was greater in rats that had received backward appetitive conditioning with the light (Protocol $\times$ Virus: $F_{(1,31)} = 23.29$; $p < 0.001$; $\eta^2 = 0.43$), as these rats displayed overall higher freezing levels than those initially trained with the tone (Protocol: $F_{(1,31)} = 20.35$; $p < 0.001$; $\eta^2 = 0.40$). The source of this difference remains unclear, and is somewhat inconsistent with the notion that stimuli predicting the omission of an appetitive outcome (the tone in this experiment) are functionally equivalent to stimuli predicting fearful events (Laurent et al., 2018 [DOI](#); Dickinson & Dearing, 1979 [DOI](#)). No significant differences were detected during the baseline period (lowest $p = 0.23$).

The test data revealed that backward *appetitive* conditioning enables IL stimulation to facilitate subsequent *fear* extinction in a stimulus-specific manner (Figure 5C [DOI](#) and Figure 5-figure supplement 1B [DOI](#)). Rats with prior backward appetitive experience with the light displayed substantial and equivalent freezing levels to the tone regardless of IL stimulation status (Diff-EYFP vs. Diff-ChR2: $p = 0.28$). These freezing levels were comparable to those observed in rats that had received backward appetitive conditioning to the tone but no IL stimulation during the brief fear extinction session (Diff-EYFP + Diff-ChR2 vs. Same-EYFP: $p = 0.98$). Critically, all three control groups exhibited significantly greater freezing than rats that had received both backward appetitive conditioning to the tone and IL stimulation during the brief fear extinction session (Diff-EYFP + Diff-ChR2 + Same-EYFP vs. Same-ChR2: $F_{(1,31)} = 6.59$; $p < 0.05$; $\eta^2 = 0.18$). No significant differences were observed during the baseline period (lowest $p = 0.26$).

These findings demonstrate that backward *appetitive* conditioning enables IL stimulation to facilitate subsequent *fear* extinction to that stimulus. This pattern suggests that inhibitory memories encoded within the IL do not contain information about the motivational properties of the omitted outcome, indicating that these memory traces are motivationally neutral.

Stimulus familiarity does not enable IL stimulation to facilitate fear extinction

The results of the previous experiments suggested that initial extinction or backward conditioning generates an inhibitory memory that can be retrieved and strengthened through IL stimulation during subsequent fear extinction. However, an alternative explanation is that the initial training session simply increased familiarity with the stimulus, and this enhanced familiarity—rather than inhibitory memory formation—enabled the facilitative effects of IL stimulation. Experiment 6 tested this alternative. Rats received the same surgical procedures ([Figure 6-figure supplement 1A](#)) and behavioral protocols as the previous experiment with one critical modification ([Figure 6A](#)). Instead of using backward appetitive conditioning to the light as the control condition, we replaced these control rats with animals that underwent forward appetitive conditioning to the tone (For-EYFP and For-ChR2). We contrasted performance of these rats with that of rats that received backward appetitive conditioning to the tone (Back-EYFP and Back-ChR2).

Appetitive data are shown in [Figure 6B](#). Forward appetitive conditioning was successful; both For-EYFP and For-ChR2 groups showed a gradual increase in magazine entries across training days (Day: $F_{(1,13)} = 61.28$; $p < 0.001$; $\eta^2 = 0.83$). Animals made significantly more entries during tone presentations compared to pre-tone periods (Period: $F_{(1,13)} = 65.68$; $p < 0.001$; $\eta^2 = 0.84$), and this discrimination became increasingly pronounced as training progressed (Period $\times$ Day: $F_{(1,13)} = 187.24$; $p < 0.001$; $\eta^2 = 0.94$). No other significant effects were observed during this phase (lowest $p = 0.15$). Backward appetitive conditioning proceeded as expected (Back-EYFP and Back-ChR2). While overall entries did not increase across days (Day: $p = 0.90$), animals showed greater entries during tone presentations than during pre-food periods (Period: $F_{(1,10)} = 97.53$; $p < 0.001$; $\eta^2 = 0.92$). This temporal discrimination pattern weakened over training sessions (Period $\times$ Day: $F_{(1,10)} = 22.86$; $p < 0.001$; $\eta^2 = 0.70$), with no other significant effects observed (lowest $p = 0.25$).

Freezing data are presented in [Figure 6C](#). Forward fear conditioning proceeded successfully, with tone-elicited freezing increasing gradually across trials (Trial: $F_{(1,23)} = 177.02$; $p < 0.001$; $\eta^2 = 0.44$). An unexpected but significant effect of viral assignment emerged ($F_{(1,23)} = 5.99$; $p < 0.05$; $\eta^2 = 0.21$), though no other significant differences were found during this stage, including during the baseline period (lowest $p = 0.07$).

The brief fear extinction session produced a gradual decline in tone-elicited freezing across trials (Trial: $F_{(1,23)} = 7.10$; $p < 0.05$; $\eta^2 = 0.25$). As expected, IL stimulation significantly reduced freezing responses (Virus: $F_{(1,23)} = 248.83$; $p < 0.001$; $\eta^2 = 0.92$). Consequently, the gradual decline in freezing was more pronounced in non-stimulated animals (Virus $\times$ Trial: $F_{(1,23)} = 8.06$; $p < 0.01$; $\eta^2 = 0.26$). Interestingly, rats that had received forward appetitive conditioning showed a more pronounced decline in freezing (Protocol $\times$ Trial: $F_{(1,23)} = 9.02$; $p < 0.01$; $\eta^2 = 0.28$), resulting in a significant three-way interaction (Protocol $\times$ Virus $\times$ Trial: $F_{(1,23)} = 5.37$; $p < 0.05$; $\eta^2 = 0.19$). While the source of these differences remains unclear, they may reflect delayed fear conditioning to the appetitive inhibitor (Back-EYFP group) combined with enhanced fear conditioning to the appetitive excitator (For-EYFP group), as suggested by previous research ([Laurent et al., 2018](#); [Dickinson & Dearing, 1979](#); [Dickinson, 1976](#)). No significant differences were observed during the baseline period (lowest $p = 0.26$).

The test data provided clear evidence that backward appetitive conditioning, not stimulus familiarity, enables IL stimulation to facilitate fear extinction ([Figure 6C](#) and [Figure 6-figure supplement 1B](#)). Rats with prior forward appetitive experience to the tone displayed substantial and equivalent freezing levels to the tone regardless of IL stimulation status (For-EYFP vs. For-ChR2: $p = 0.40$). These freezing levels were comparable to those observed in rats that had received backward appetitive conditioning to the tone but no IL stimulation during the brief fear extinction session (For-EYFP + For-ChR2 vs. Back-EYFP: $p = 0.11$). Critically, all three control groups exhibited significantly greater freezing than rats that had received both backward appetitive conditioning to the tone and IL stimulation during the brief fear extinction session (For-EYFP + For-ChR2 + Back-EYFP vs. Back-ChR2: $F_{(1,31)} = 6.81$; $p < 0.05$; $\eta^2 = 0.23$). No significant differences were observed during the baseline period (lowest $p = 0.41$).

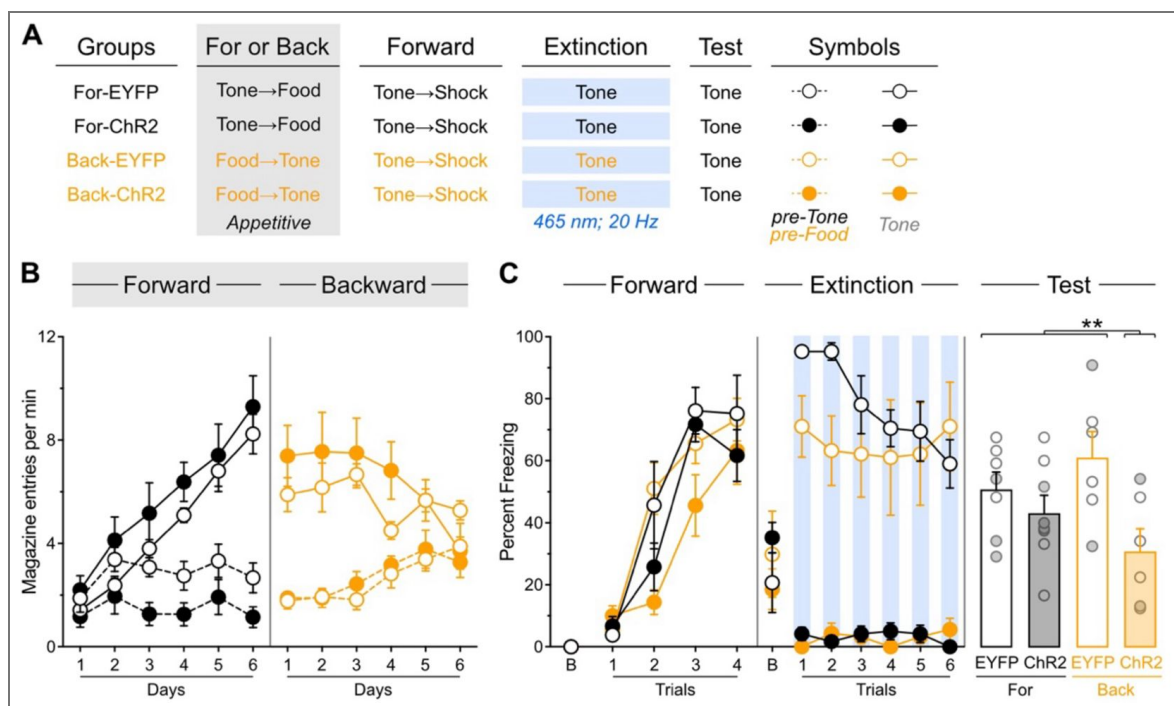


Figure 6. Stimulus familiarity does not enable IL stimulation to facilitate fear extinction

A. Schematic representation of the behavioral design used in Experiment 6. The LEDs were activated in all groups during the fear extinction session. For-EYFP: $n = 3$ females and 4 males; For-ChR2: $n = 4$ females and 4 males; Diff-EYFP: $n = 2$ females and 4 males; Diff-ChR2: $n = 2$ females and 4 males. **B.** Number of magazine entries during forward and backward appetitive conditioning. **C.** Mean percent freezing during remaining experimental stages. Data are shown as mean $\pm$ SEM. Test data include individual data points for female (filled circle) and male (open circle) rats. Asterisks denote significant effect (** $p < 0.01$).

These findings demonstrate that stimulus familiarity per se does not enable IL stimulation to facilitate fear extinction. Instead, the results support our original hypothesis: initial extinction or backward conditioning generates a specific inhibitory memory that can be retrieved and strengthened by IL stimulation during subsequent fear extinction to the same stimulus.

Prior experience with appetitive extinction enables IL stimulation to facilitate subsequent backward fear conditioning

Experiment 7 investigated whether the facilitative effects of IL stimulation are limited to fear extinction sessions or can be observed when stimulation occurs during other types of learning that involve inhibition. Specifically, we examined whether these facilitative effects could also be uncovered during backward fear conditioning. Rats received bilateral IL infusions of either EYFP or ChR2 virus and were implanted with fiber-optic cannulae targeting the IL (Figure 7-figure supplement 1A [↗](#)). All animals underwent the same behavioral protocol (Figure 7A [↗](#)) using the tone as the only stimulus. The sequence of behavioral stages was: forward appetitive conditioning, appetitive extinction, forward fear conditioning, backward fear conditioning, and test. LED activation occurred during 6 of the 8 tone presentations during the backward fear conditioning phase.

Appetitive data are shown in Figure 7B [↗](#). Forward appetitive conditioning was successful, with overall entries increasing gradually across days (Day: $F_{(1,17)} = 106.67$; $p < 0.001$; $\eta^2 = 0.86$). Animals made significantly more entries during tone presentations compared to pre-tone periods (Period: $F_{(1,17)} = 19.50$; $p < 0.001$; $\eta^2 = 0.53$), and this discrimination became increasingly pronounced as training progressed (Period $\times$ Day: $F_{(1,13)} = 144.51$; $p < 0.001$; $\eta^2 = 0.89$). No other significant effects were observed during this phase (lowest $p = 0.15$). Appetitive extinction produced a gradual decrease in overall entries (Trial: $F_{(1,17)} = 105.63$; $p < 0.001$; $\eta^2 = 0.92$). Although entries were greater during tone presentations compared to pre-tone periods (Period: $F_{(1,17)} = 55.67$; $p < 0.001$; $\eta^2 = 0.77$), this discrimination became progressively smaller (Period $\times$ Trial: $F_{(1,17)} = 35.53$; $p < 0.001$; $\eta^2 = 0.68$).

Freezing data are presented in Figure 7C [↗](#). Forward fear conditioning proceeded successfully, with tone-elicited freezing increasing across trials (Trial: $F_{(1,17)} = 52.83$; $p < 0.001$; $\eta^2 = 0.76$) regardless of viral assignment ($p = 0.95$). No other significant differences were found during this stage, including during the baseline period (lowest $p = 0.026$). Backward fear conditioning proceeded as expected, with overall freezing greater during the pre-shock periods compared to stimulus presentations (Period: $F_{(1,17)} = 43.54$; $p < 0.001$; $\eta^2 = 0.72$). Critically, tone-elicited freezing was reduced by LED activation (LED: $F_{(1,17)} = 11.98$; $p < 0.01$; $\eta^2 = 0.41$), but this reduction was only observed in ChR2 rats (LED $\times$ Virus: $F_{(1,17)} = 22.25$; $p < 0.001$; $\eta^2 = 0.57$). No other significant differences were observed during this stage (lowest $p = 0.06$).

The test data revealed that IL stimulation during backward fear conditioning successfully reduced subsequent freezing (Figure 7C [↗](#) and Figure 7-figure supplement 1B [↗](#)): ChR2 rats froze less than EYFP rats (Virus: $F_{(1,17)} = 6.67$; $p < 0.05$; $\eta^2 = 0.28$). This finding demonstrates that appetitive extinction enables IL stimulation to facilitate subsequent backward fear conditioning. This indicates that the facilitative effects of IL stimulation are not limited to fear extinction sessions but can be observed when stimulation occurs during other procedures that involve inhibitory learning. This pattern suggests that procedures generating inhibition create a memory within the IL that can be accessed during other inhibitory learning situations.

Discussion

The present experiments explored how the IL encodes inhibitory memories by investigating the conditions that promote their retrieval. Initial ex-vivo electrophysiological recordings confirmed that optogenetic stimulation successfully increased activity in IL neurons. We then found that prior experience with fear extinction enables IL stimulation to facilitate subsequent fear extinction. The facilitative effects of IL stimulation were not limited to fear extinction experience, as prior exposure to backward fear conditioning also enabled IL stimulation to enhance later fear

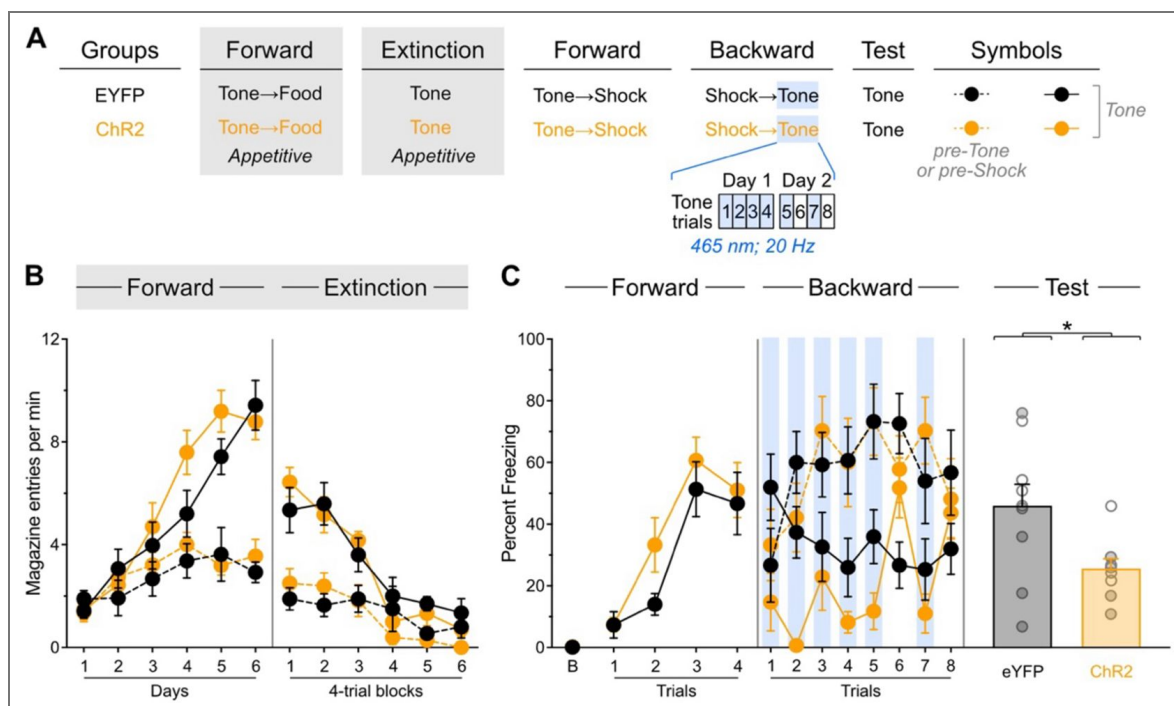


Figure 7. Prior experience with appetitive extinction enables IL stimulation to facilitate subsequent backward fear conditioning

A. Schematic representation of the behavioral design used in Experiment 7. The LEDs were activated in all groups during backward fear conditioning. EYFP: $n = 5$ females and 5 males; ChR2: $n = 5$ females and 4 males. **B.** Number of magazine entries during forward appetitive conditioning and appetitive extinction. **C.** Mean percent freezing during remaining experimental stages. Data are shown as mean $\pm$ SEM. Test data include individual data points for female (filled circle) and male (open circle) rats. Asterisks denote significant effect ($* p < 0.05$).

extinction. These effects were stimulus-specific, occurring only for the stimulus that had undergone backward fear conditioning, a procedure that was found to generate the characteristics of inhibitory learning. Remarkably, backward appetitive conditioning also allowed IL stimulation to facilitate later fear extinction, suggesting that inhibitory memories in the IL lack information about the motivational circumstances present during their encoding. Additional experiments ruled out stimulus familiarity as an explanation for the facilitative effects of IL stimulation and demonstrated that IL-mediated facilitation can be observed in procedures other than fear extinction. Together, these findings provide novel insights into how the brain processes and stores inhibitory information and broaden our understanding of IL function in adaptive behavior.

Throughout all experiments, IL stimulation produced an immediate suppression of the freezing responses elicited by the conditioned stimulus. This effect aligns with the existing literature (Do-Monte et al., 2015 [DOI](#); Kim et al., 2016 [DOI](#); Milad et al., 2004 [DOI](#); Vidal-Gonzalez et al., 2006 [DOI](#); Lingawi et al., 2018 [DOI](#); Lingawi et al., 2017 [DOI](#); Thompson et al., 2010 [DOI](#); Chang & Maren, 2011 [DOI](#)), but it did not predict the long-term consequences of the stimulation. This dissociation between immediate and long-term effects was evident in Experiment 1. While IL stimulation suppressed fear responses during both the first and second extinction sessions, it only facilitated the fear inhibition produced by the second session, not the first. This result confirmed that prior experience with fear extinction creates the conditions necessary for IL stimulation to enhance subsequent extinction learning (Chang & Maren, 2011 [DOI](#); Lingawi et al., 2017 [DOI](#)). Critically, the finding in Experiment 2 that backward fear conditioning experience also enabled IL stimulation to facilitate subsequent fear extinction demonstrated that IL engagement extends beyond situations where stimuli are encountered in isolation (e.g., extinction) to encompasses protocols believed to generate inhibitory learning through different mechanisms (Sosa, 2024 [DOI](#)). This finding was replicated in Experiment 3, which also showed that the facilitative effects of IL stimulation were not due to background context fear and only emerged when the same stimulus undergoes both backward fear conditioning and subsequent fear extinction, indicating that backward fear conditioning generates stimulus-specific memories within the IL. These memories were found to be inhibitory, as Experiments 2 and 3 revealed retarded fear conditioning to the backward-trained stimulus, while Experiment 4 found that such a stimulus reduced the fear-eliciting capacity of a separately fear-conditioned stimulus, both of which are hallmark properties of inhibitory stimuli (Rescorla, 1969 [DOI](#)). Collectively, these findings support the hypothesis that fear extinction and backward fear conditioning generate stimulus-specific inhibitory memories that can be subsequently reactivated during a fear extinction session to facilitate the inhibition produced by that session.

Experiment 5 asked about the motivational characteristics of inhibitory memories encoded within the IL. It found that prior experience with *appetitive* backward conditioning enabled IL stimulation to facilitate *fear* extinction in a stimulus-specific manner. This suggests that inhibitory memories stored in the IL are stripped of the motivational information that is present during their initial encoding, a suggestion that is consistent with the earlier observation that *appetitive* extinction enables IL stimulation to facilitate subsequent *fear* extinction (Lingawi et al., 2018 [DOI](#)). Experiment 6 addressed the alternative hypothesis that IL stimulation became effective in all experiments simply because prior experiences enhanced stimulus familiarity. The results demonstrated that prior experience with backward appetitive conditioning, but not forward appetitive conditioning, enabled IL stimulation to facilitate subsequent fear extinction. Since the backward and forward protocols provided similar exposure to the target stimulus, this finding rules out familiarity as the determining factor in IL engagement and supports the view that the critical determinant is the capacity of prior experiences to produce inhibitory learning. Finally, Experiment 7 sought to demonstrate that IL-encoded inhibitory memories can be retrieved during inhibitory procedures other than fear extinction, thereby expanding the scope of IL function beyond that of fear extinction. It found that prior experience with appetitive extinction enabled IL stimulation to facilitate the development of fear inhibition during backward fear conditioning. This finding generalizes the effects of IL stimulation to another inhibitory protocol, providing further evidence that the IL generally contributes to inhibitory learning even in different motivational contexts.

The present experiments aimed to uncover the general role of the IL in encoding inhibitory memories. Accordingly, they employed a non-selective optogenetic approach that stimulated all IL neurons. The results reveal an important difference with studies that employed cell-specific stimulation. Although selective stimulation of IL pyramidal neurons has been found to facilitate initial extinction learning (Do-Monte et al., 2015 [DOI](#); Chen et al., 2021 [DOI](#)), this facilitation was absent in the present experiments, which instead replicated previous work that used a pharmacological IL stimulation approach. These contrasting effects extend previous findings showing that extinction retrieval is disrupted by broad optogenetic silencing of the IL (Kim et al., 2016 [DOI](#)) but is left unaffected by selective silencing of IL pyramidal neurons (Do-Monte et al., 2015 [DOI](#)). Together, these discrepancies highlight the critical role that IL interneurons may play in shaping both the encoding and expression of inhibitory memories, which is consistent with recent research uncovering an important function for these interneurons in extinction learning (Ho et al., 2025 [DOI](#); Marek et al., 2018a [DOI](#); Binette et al., 2023 [DOI](#); Koppensteiner et al., 2019 [DOI](#)). Future studies should therefore identify the specific neuronal populations that are responsible for the effects observed in the present experiments. Additionally, it will be necessary to examine how these effects are modulated by other brain regions that have been identified as important regulator of inhibitory learning. These include the amygdala (Do-Monte et al., 2015 [DOI](#); Laurent et al., 2008 [DOI](#); Laurent & Westbrook, 2008 [DOI](#)), the hippocampus (Hassell et al., 2025 [DOI](#); Marek et al., 2018a [DOI](#)), the thalamus, locus coeruleus (Bayer et al., 2025 [DOI](#); Do-Monte et al., 2010 [DOI](#); Mueller et al., 2008 [DOI](#); Uematsu et al., 2017 [DOI](#)), the orbitofrontal cortex (Lay et al., 2020 [DOI](#); Rodriguez-Romaguera et al., 2015 [DOI](#)), and the lateral habenula (Matsumoto & Hikosaka, 2007 [DOI](#); Laurent et al., 2017 [DOI](#); Sosa et al., 2021 [DOI](#)).

Other aspects of the present data merit consideration. First, despite our efforts to match group performance during training, post-mortem exclusions resulted in unexpected group differences in some experiments. However, these differences were typically transient or unable to account for the final experiment outcomes. Moreover, the key findings presented here are replicated across several experiments within the current study (e.g., prior experience with backward fear or appetitive conditioning) or reproduced previous work (prior experience with fear or appetitive extinction; Lingawi et al., 2018 [DOI](#); Lingawi et al., 2017 [DOI](#)). Therefore, considerable evidence attest to the reliability and replicability of the findings reported. A second consideration relates to the mechanisms underlying the IL facilitatory effects observed here. Future research will be needed to determine whether IL stimulation enhances the subsequent retrieval of inhibitory memories or accelerates the relearning of these memories, as either process could account for the improved performance observed in our experiments. Finally, additional studies will be needed to establish whether the characteristics of IL function described here apply equally to both sexes. Although our experiments included both female and male rats, they lacked sufficient statistical power to detect sex differences. Given the substantial evidence for sex differences in extinction learning and related behaviors (Tang & Graham, 2020 [DOI](#)), this remains an important question for future investigation.

Collectively, the present findings reveal fundamental principles governing how the IL encodes and retrieves inhibitory memories. Our experiments demonstrate that the IL stores inhibitory memories that are extremely flexible since they can be retrieved and used across many inhibitory procedures and distinct motivational contexts. This motivational independence offers a sophisticated neural mechanism that allows the application of previously learned inhibitory relationships to novel circumstances. Whether inhibitory memories operate entirely outside motivational control or whether this autonomy specifically characterizes the components that are encoded in the IL remains unknown, though previous work showing that inhibitory stimuli can modulate learning and behavior regardless of primary motivational states suggests that inhibitory memories may broadly escape motivational regulation (Laurent et al., 2018 [DOI](#)). Regardless, the stimulus specificity of the reported effects reveal precision in IL memory encoding, restricting retrieval to appropriate situations, and balancing for the generalizability of retrieval provided its motivational independence. By ruling out alternative explanations such as context fear or stimulus familiarity, our experiments also provide evidence that the IL is primarily concerned

with inhibitory learning rather than with more basic associative processes. Thus, the IL emerges as a critical hub for the flexible application of inhibitory knowledge that enables adaptive responses in the face of ever-changing environmental circumstances.

Materials and methods

Subjects

Subjects were 248 experimentally naïve female and male Long-Evans rats obtained from the breeding facility at the University of New South Wales (Sydney, Australia). The rats were at least 8 weeks old and weighed 300-500 grams at the beginning of each experiment. Efforts were made to match group performance across the training stages and to allocate an equal number of female and male rats in each group. However, these efforts were hampered by animal's availability and the necessary exclusions following post-mortem assessments of viral spread and fiber-optic cannula placement. The final number of female and male rats did not provide sufficient power to analyze an effect of sex. Therefore, all test data present individual data for female and male rats. Rats were housed in transparent intraventilated cages (Tecniplast Australia) with their littermates (up to four rats per cage) in a climate-controlled colony room maintained on a 12-hour light-dark cycle (lights on at 7:00 am). Experimental procedures were conducted during the light phase (8:00am to 6:00 pm). Water and standard lab chow were available *ad libitum* except in experiments involving the delivery of food outcomes. In these experiments, rats were food restricted to maintain them at ~90% of their *ad libitum* body weight. Food restriction started 3-4 days before the start of the experiments and was maintained throughout the behavioral procedures. All rats were handled daily 3-4 days before the start of the experiments. All the animals were handled according to approved Animal Care and Ethics Committee (ACEC) protocols of the University of New South Wales. The experimental protocols were approved by the UNSW ACEC (Permit Number: 21/28B). All surgery was performed under isoflurane anesthesia, and every effort was made to minimize suffering.

Viruses

The following viruses were used: AAV5-hSyn-eYFP (**EYFP**; University of North Carolina Vector Core; Chapel Hill, NC, USA) and AAV5-hSyn-hChR2(H134R)-EYFP (**ChR2**; Addgene; Watertown, MA, USA; #26973).

Surgery

Stereotaxic surgery was conducted under isoflurane gas anesthesia (0.8L/min; induction at 4-5%, maintenance 2-2.5%). Animals were placed in a stereotaxic frame (Kopf Instruments, Tujunga, CA, USA), the surgical area was shaved, and betadine iodine antiseptic solution was applied. At the incision site, bupivacaine hydrochloride (0.5%; 0.1-0.2 mL) was injected subcutaneously. In the lower flank, Metacam (1 mg/kg) was injected subcutaneously. An incision was made along the midline to expose the skull, the membrane on top of the skull was cleared, and the skull was adjusted to align bregma and lambda on the same horizontal plane. Small holes were drilled into the skull above the infralimbic cortex (IL) using the following coordinates A/P: +2.45, M/L: ±2.0, D/V: -5.1 at 15° angle. An Infuse/Withdraw Pump (Standard Infuse/Withdraw Pump 11 Elite; Harvard Apparatus, Holliston, MA, USA) in combination with glass 1 µL syringes (86200; Hamilton Company, Reno, NV, USA) were used to perform viral infusions. 0.5 µL of the virus (EYFP or ChR2) was infused in each hemisphere at a rate of 0.1 µL/min. Following each infusion, the needle was left in position for 5-10 min to allow for diffusion. Fiber-optic cannulae were then bilaterally implanted 0.5 mm above and fixed with jeweler's screws and dental cement. Animals then received intraperitoneal injections of sodium chloride (0.9%, 2 mL) and procaine penicillin (Ilium Benicillin; 300mg/kg) were placed on a heat mat for recovery. Once alert and responsive, animals were returned to their home cages and monitored daily as they recovered for 3 weeks to allow for viral expression before behavioral procedures commenced.

Behavioral apparatus

Training and testing were conducted in 8 identical operant chambers (ENV-007-VP; L 29.53 x W 23.5 x H 27.31 cm; MED Associates, Fairfax, VT, USA) enclosed within light- and sound-attenuating cabinets (ENV-018MD; L 59.69 x W 40.64 x H 55.88 cm; MED Associates, Fairfax, VT, USA). The chambers consisted of two stainless-steel side walls, three clear polycarbonate walls and ceiling, and a stainless-steel grid floor. Each chamber was equipped with a speaker that delivered a sound card-generated (ANL-926; MED Associates, Fairfax, VT, USA) 90 dB, 1000 Hz tone. LED lights mounted on the back wall of each sound-resistant cabinet were used to deliver a flashing light stimulus presented at 2 Hz (~8 lx). The duration of the tone and flashing stimuli was set at 30 seconds in all behavioral procedures. The grid floor of each chamber was connected to a shock generator (ENV-410C; MED Associates, Fairfax, VT, USA) that delivered a 0.5 mA, 0.5s shock. Grain pellets (F0165; Bio-Serv Biotechnologies, Flemington, NJ, USA) could be delivered through a food magazine in one wall of the chamber. Head entries into the magazine were detected via an infrared beam that crossed the magazine opening. The ceiling of the chambers contained a hole that enabled the connection of the patch cable between the LED and each freely moving animal. Training and testing sessions were programmed and controlled by computers external to the testing rooms using MED-PC V software (MED Associates, Fairfax, VT, USA) which also recorded experimental data from each session. All cabinets were fitted with cameras (IPC-K35AP; Dahua, Artarmon, NSW, Australia) to record and view real-time chamber activity.

Optogenetic equipment

Fiber-optic cannulas (10mm; RWD Life Science Co., Guangdong, China) consisted of a ceramic ferrule from which a silica/polymer flat-tip 400 nm core fibre extended. Fiber-optic patch cords (custom; Doric Lenses, Québec, QC, Canada) were enclosed in a flexible cladding that consisted of one transparent fiber that split into two smaller fibres for attachment to the implanted fiber-optic cannulae. Delivery of the LED light was controlled through a TTL adapter which converted 28V DC output to a TTL transition (MED Associates, Fairfax, VT, USA). The adaptor was connected to an LED driver (Doric Lenses, Québec, QC, Canada) and Connectorized LED light source (Doric Lenses, Québec, QC, Canada) with a fiber-optic rotary joint for attachment to the fiber-optic patch cords. LED light was delivered as blue light (465 nm) and measured at least 10mW at the cannula tip using a photometer (PM200; ThorLabs, Newton, NJ, USA) when the LED was switched on.

Behavioral procedures

All experiments employed parameters similar to those from our previous studies (Laurent et al., 2022 [DOI](#); Lingawi et al., 2018 [DOI](#); Laurent et al., 2018 [DOI](#); Lingawi et al., 2017 [DOI](#)).

Experiment 1

Day 1: Pre-exposure

All rats received pre-exposure to the experimental chambers and tone stimulus during a 30-minute session. The tone was presented twice: first after 11 minutes, then again 11 minutes later, with rats removed 7 minutes after the second presentation. Patch cords were connected to fiber-optic cannulae and LEDs throughout this session, though LEDs remained inactive.

Day 2: Forward fear conditioning

All rats underwent forward fear conditioning with four tone-shock pairings. The tone stimulus co-terminated with a single shock, with pairings separated by pseudorandom inter-trial intervals (ITI) of 3-5 minutes (4-minute average). Total session duration was 20 minutes.

Day 3: Standard fear extinction

Rats in the ReExt-OFF and ReExt-ON groups received their first extinction session, consisting of 16 tone-alone presentations with variable ITI of 3-5 minutes over 58 minutes. Rats in the Ext-OFF and Ext-ON groups were handled only.

Day 4: Second fear conditioning

All rats received an additional forward fear conditioning session.

Day 5: Brief fear extinction

All rats underwent a shortened extinction session with 8 tone-alone presentations and variable ITI of 3-5 minutes, lasting 38 minutes total. This represented the first extinction session for Ext-OFF and Ext-ON groups, but the second for ReExt-OFF and ReExt-ON groups. Critically, patch cords were attached to fiber-optic cannulae and LEDs during this session, with LEDs activated during tone presentations for the Ext-ON and ReExt-ON groups only.

Day 6: Test

All rats received 8 tone-alone presentations with variable ITIs of 3-5 minutes over a 38-minute session.

Experiment 2

Day 1: Pre-exposure

All rats underwent pre-exposure sessions using previously described methods.

Days 2-6: Backward fear conditioning

Rats in the Back-EYFP and Back-ChR2 groups received 5 daily backward fear conditioning sessions with the tone stimulus. Each session consisted of 4 shock-tone trials, where shocks were delivered 5 seconds before tone presentation. Variable ITI of 3-5 minutes were used, with each session lasting 20.5 minutes. Rats in the No-EYFP and No-ChR2 groups were handled only during this period.

Days 7-8: Context extinction

All rats underwent 2 daily 30-minute exposure sessions in the conditioning chambers without stimulus presentation. During these sessions, rats were habituated to patch cord attachment using the pre-exposure protocol.

Day 9: Forward fear conditioning

All rats received forward fear conditioning to the tone using previously described methods.

Day 10: Brief fear extinction

All rats underwent a shortened extinction session to the tone following previously described protocols. Patch cords were connected to fiber-optic cannulae and LEDs throughout this session, with LEDs activated during tone presentations for all experimental groups.

Day 11: Test

All rats were tested for conditioned fear responses to the tone using previously described procedures.

Experiment 3

Experiment 3 followed the same protocol as Experiment 2, with one key modification: control groups (Diff-EYFP and Diff-ChR2) received backward fear conditioning to a light stimulus, while experimental groups (Same-EYFP and Same-ChR2) received backward fear conditioning to the tone stimulus.

Experiment 4

Days 2-6: Backward fear conditioning

The Back group received backward fear conditioning to the tone stimulus following previously described methods. The No group underwent identical procedures with tone presentations excluded.

Days 7-8: Context extinction

All rats received 2 daily 30-minute sessions of chamber exposure without stimulus presentation.

Day 9: Forward fear conditioning

All rats underwent forward fear conditioning using previously described protocols, substituting the light stimulus for the tone stimulus.

Day 10: Test

All rats were tested for conditioned fear responses to both the light alone and the light-tone compound stimulus. Each stimulus type was presented 4 times in pseudorandom order (ABBABAAB; with A = light and B = light + tone) with variable ITI of 3-5 minutes over a 38-minute session.

Experiment 5

Experiment 5 followed the same protocol as Experiment 3, with backward appetitive conditioning replacing backward fear conditioning.

Days 2-7: Backward appetitive conditioning

Over six daily sessions, rats received conditioning with either the light stimulus (Diff-EYFP and Diff-ChR2 groups) or the tone stimulus (Same-EYFP and Same-ChR2 groups). Each session consisted of 6 trials with variable ITIs of 3-5 minutes, lasting approximately 60 minutes total. Each trial began with the delivery of two grain pellets. Upon detection of a food magazine entry, the light or tone stimulus was activated 5 seconds later and continued for 30 seconds before terminating.

Experiment 6

Experiment 6 followed the same protocol as Experiment 5, with one key modification: on days 2-7, control rats (For-EYFP and For-ChR2 groups) received forward appetitive conditioning with the tone stimulus instead of backward appetitive conditioning with the light stimulus.

Days 2-7: Forward appetitive conditioning

The same parameters as backward appetitive conditioning were used, except that two grain pellets were delivered immediately after tone stimulus presentation.

Experiment 7

Rats underwent the following sequence: pre-exposure, forward appetitive conditioning, appetitive extinction, forward fear conditioning, backward fear conditioning, and a final test. All stages employed the tone stimulus using previously described parameters with minor modifications: backward fear conditioning occurred over only two days and was conducted with LED activation during all tone presentations on the first day but only during the first and third tone presentation on the second day. Appetitive extinction involved 2 daily sessions with 15 tone-alone presentations and variable ITIs of 3-5 minutes.

Tissue preparation, histology and microscopy

Rats were euthanised with an intraperitoneal injection of sodium pentobarbital (1mL) and transcardially perfused with 4% paraformaldehyde (PFA) in 0.1M sodium phosphate buffer (PBS, pH 7.5; 400mL for rats). Brains were immediately removed and placed in individual specimen jars containing PFA and stored at 4°C overnight for post-fixation. Brains were sliced into 40µm coronal or sagittal sections in 0.1M Tris-buffered saline (TBS, 0.25M Tris, 0.5M NaCl, pH 7.5) using a vibratome (VT1200; Leica Microsystems, North Ryde, NSW, Australia) and stored at -20°C in cryoprotective solution containing 30% ethylene glycol, 30% glycerol and 0.1M PBS. Free-floating sections were selected in TBS to contain the region of interest and washed three times for 10 min each in TBS solution. Next, sections were mounted onto glass slides using Vectashield hardset antifade mounting medium (H-1400; Vector Laboratories, Newark, CA, USA).

Imaging of the mounted brain sections was performed on a confocal microscope (BX61W1 or FV10i; Olympus, Shinjuku, Tokyo, Japan) using Fluoview software (FV1000; Olympus, Shinjuku, Tokyo, Japan). Acquired images were used to assess the spread of viral infusions and placement of

fiber-optic cannulae. Based on the latter, the following numbers of rats were excluded from the statistical analyses: Experiment 1 ($n = 3$), Experiment 2 ($n = 11$), Experiment 3 ($n = 6$), Experiment 5 ($n = 3$), Experiment 6 ($n = 3$) and Experiment 7 ($n = 3$).

Electrophysiological recordings

Brain slice preparation

Rats were euthanised under isoflurane gas anaesthesia (4% in air) four weeks following bilateral IL infusion of the ChR2 virus. Brains were immediately removed and sliced using a vibratome (VT1200; Leica Microsystems, North Ryde, NSW, Australia) in ice-cold oxygenated sucrose buffer solution (consisting of [in mM]: 241 sucrose, 28 NaHCO_3 , 11 glucose, 1.4 NaH_2PO_4 , 3.3 KCl, 0.2 CaCl_2 , 7 MgCl_2). Coronal brain slices (300 μm thickness) containing the IL were sampled and maintained at 33°C in a submerged chamber containing physiological saline (consisting of [in mM]: 124 NaCl, 2.5 KCl, 1.25 NaH_2PO_4 , 1 MgCl_2 , 1 CaCl_2 , 10 glucose and 26 NaHCO_3) and equilibrated with 95% O_2 and 5% CO_2 .

Recordings

After equilibration for 1 h, slices were transferred to a recording chamber and neurons visualized under an upright microscope (BX50WI, Olympus) using differential interference contrast (DIC) Dodt tube optics and EYFP fluorescence, and superfused continuously (1.5 ml/min) with oxygenated physiological saline at 33°C. Cell-attached and whole-cell patch-clamp recordings were made using electrodes (2–5 MW) containing internal solution (in mM): 115 K gluconate, 20 NaCl, 1 MgCl_2 , 10 HEPES, 11 EGTA, 5 Mg-ATP, and 0.33 Na-GTP, 5 phosphocreatine di(tris) salt (#P1937, Sigma-Aldrich), pH 7.3, osmolarity 285–290 mOsm/L. Biocytin (0.1%; #B4261, Sigma) was added to the internal solution for marking the sampled neurons during recording. Data acquisition was performed with a Multiclamp 700B amplifier (Molecular Devices), connected to a PC computer and interface ITC-18 (Instrutech). Liquid junction potentials of -10 mV were not corrected. In cell-attached and then whole-cell current-clamp modes, membrane current and potentials were sampled at 5 kHz (low pass filter 2 kHz, Axograph X, Axograph).

LED stimulation

Transfected neurons with ChR2 in the IL were illuminated by a pulse train of LED light (470 nm, 5 ms pulse, 20 Hz, 0.5 mW, for 31 s train in cell-attached mode, and then for 2.5 s train in whole-cell mode (ThorLabs, Newton, NJ, USA) onto the brain slice under the 40X water-immersion objective.

Post-hoc histological analysis

Immediately after physiological recording, brain slices containing biocytin-filled neurons were fixed overnight in 4% paraformaldehyde/0.16 M phosphate buffer (PB) solution, rinsed and then placed in 0.5% Triton X-100/PB for 3 days to permeabilize cells. Slices were then incubated in primary antibody for 3 days, chicken anti-GFP (GFP-1020, 1:1000, Aves Labs) to enhance EYFP signal, plus 2% horse serum and 0.2% Triton X-100/PB at 4°C. After rinse, this was followed by a one-step overnight incubation at 4°C in a fluorescent avidin plus a secondary antibody, ExtrAvidin Cy3 (E4142, 1:1000, Sigma), and donkey anti-chicken Alexa 488 conjugated IgY (703-545-155, 1:1000, Jackson ImmunoResearch Inc). Stained slices were rinsed, mounted onto glass slides, dried, and coverslipped with Fluoromount-G mounting medium (00-4958-02, Invitrogen). A 2D projection was later obtained from a collated image stack using confocal laser scanning microscopy (Fluoview FV1000, BX61WI microscope, Olympus).

Statistical analyses

Two rats from Experiment 1 (group ReExt-ON) were excluded from the final analyses because they failed to display freezing during the first forward fear conditioning and extinction stages. The data presented met the assumptions of the statistical test used. The main measures were percent freezing and number of magazine entries per minute. Freezing was rated in a time-sampling manner and judged as either freezing or not freezing every 2s by a trained observer blind to the subjects' group assignment. Percent freezing was calculated for three periods: baseline was the

period prior to the onset of the first stimulus presentation, pre-shock was the period during the 30 seconds prior to the onset of each shock during backward fear conditioning, and stimulus was the period during the 30-second stimuli (tone, light or light+tone). The number of magazine entries per minute was also calculated for three periods: pre-stimulus was the period during the 30 seconds prior to the onset of each stimulus presentation, pre-food was the period during the 30 seconds prior to the onset of each 2-pellet delivery during backward appetitive conditioning, and stimulus was the period during the 30-second stimuli (tone or light).

The difference between groups were analyzed using a planned orthogonal contrast procedure controlling the per contrast error rate (Hays, 1963 [\[1\]](#)). We first consider the 4- group design experiments (Experiments 1, 2, 3, 5 and 6). All training stages included a first contrast for behavioral procedure (e.g., Ext vs. ReExt or No vs. Back), a second contrast for either LED (ON vs. OFF in Experiment 1) or Virus (EYFP vs. Chr2) and a third contrast testing for the interaction between these two contrasts. For the test stage, the contrasts were informed by our a priori hypotheses. The first contrast tested for differences between the two behavioral control groups (e.g., Ext-EYFP vs. Ext-ChR2, or No-EYFP vs. No-ChR2). The second contrast tested for differences between the two previous control groups and the behavioral experimental group in the control LED or Virus condition (e.g., ReExt-OFF, or Back-EYFP). The final contrast tested for differences between the previous three groups and the critical behavioral group in the LED ON or Virus Chr2 condition. A single between-subject contrast was used in the 2-group design experiments (Experiments 4 and 7). In all experiments, a within-subject factor assessed changes across days, trials, or block of trials using planned linear analyses. When using days, the average responding for each relevant Period (see previous paragraph) in each group was used. Trials or block of trials was used for forward fear conditioning, fear and appetitive extinction, test and backward conditioning in the last experiment. Experiment 4 employed a within-subject factor comparing freezing during light and light+tone.

All analyses were carried out using the PSY statistical software (School of Psychology, the University of New South Wales, Australia). The Type I error rate was controlled at $\alpha = 0.05$ for each contrast tested. If interactions were detected, follow-up simple effects analyses were calculated to determine the source of the interactions. For each significant statistical comparison, we report measures of effect size partial eta-squared (η^2 ; $\eta^2 = 0.01$ is a small effect, $\eta^2 = 0.06$ a medium effect and $\eta^2 = 0.14$ a large effect).

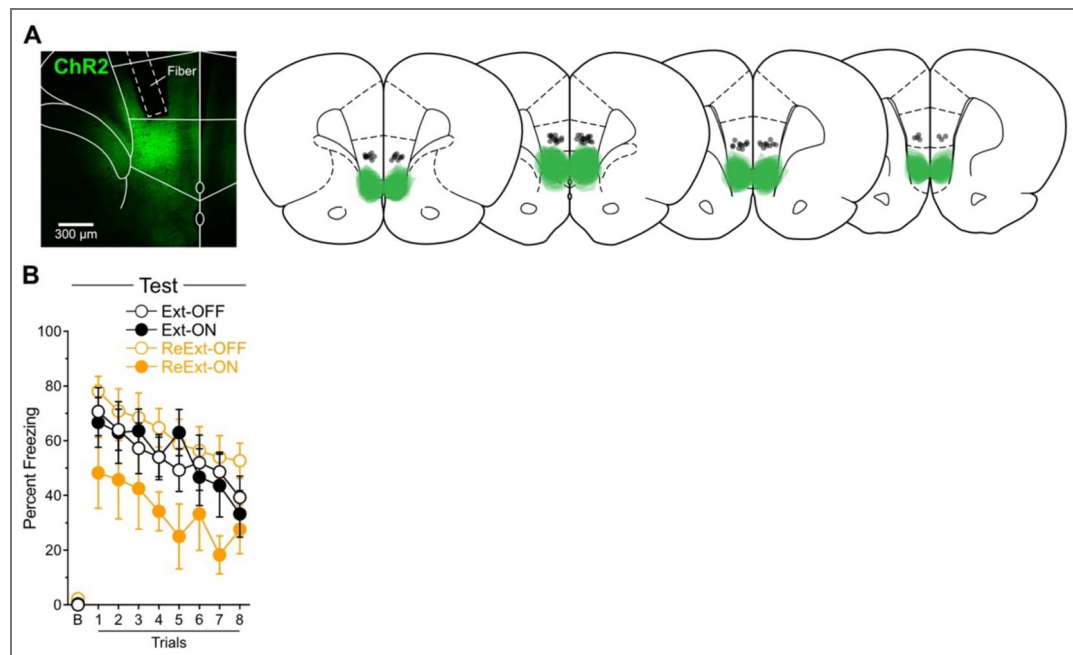


Figure 2 – figure supplemental 1. Histological and behavioral data related to Figure 2B. **A.** Data for Experiment 1. Micrographs showing ChR2 expression in IL neurons. Minimal (light green) and maximal (darker green) extent of IL viral expression. Location of fiber-optic cannulas is also represented (black circles). From left to right: +3.72; +3.24; +3.00; +2.76 mm from bregma. **B.** Freezing across trials during the test presented in Figure 2B. Freezing decreased across trials (Trial: $F_{(1,34)} = 24.20$; $p < 0.001$; $\eta^2 = 0.42$), but this decrease did not interact with the planned orthogonal contrasts reported in the manuscript (lowest $p = 0.77$). Data are shown as mean $\pm$ SEM.

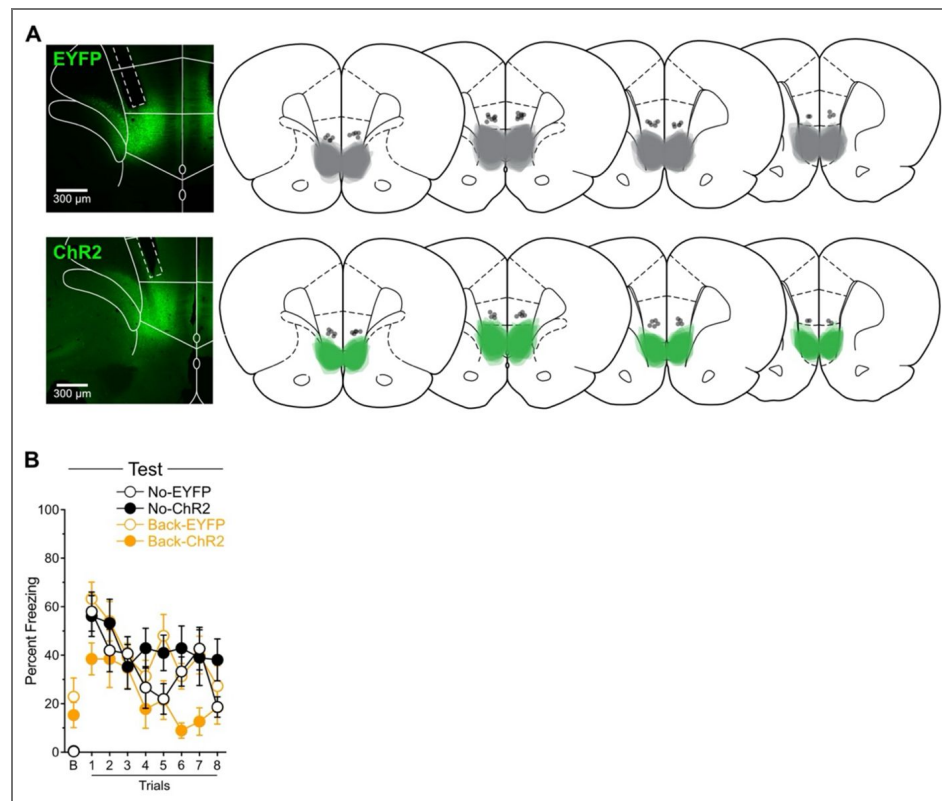


Figure 3 – figure supplemental 1. Histological and behavioral data related to Figure 3.

A. Data for Experiment 2. Micrographs showing EYFP and ChR2 expression in IL neurons. Minimal (light grey for EYFP and light green for ChR2) and maximal (dark grey for EYFP and dark green for ChR2) extent of IL viral expression. Location of fiber-optic cannulas is also represented (black circles). From left to right: +3.72; +3.24; +3.00; +2.76 mm from bregma. **B.** Freezing across trials during the test presented in Figure 3B. Freezing decreased across trials (Trial: $F_{(1,32)} = 31.53$; $p < 0.001$; $\eta^2 = 0.50$), but this decrease did not interact with the planned orthogonal contrasts reported in the main manuscript (lowest $p = 0.42$). Data are shown as mean $\pm$ SEM.

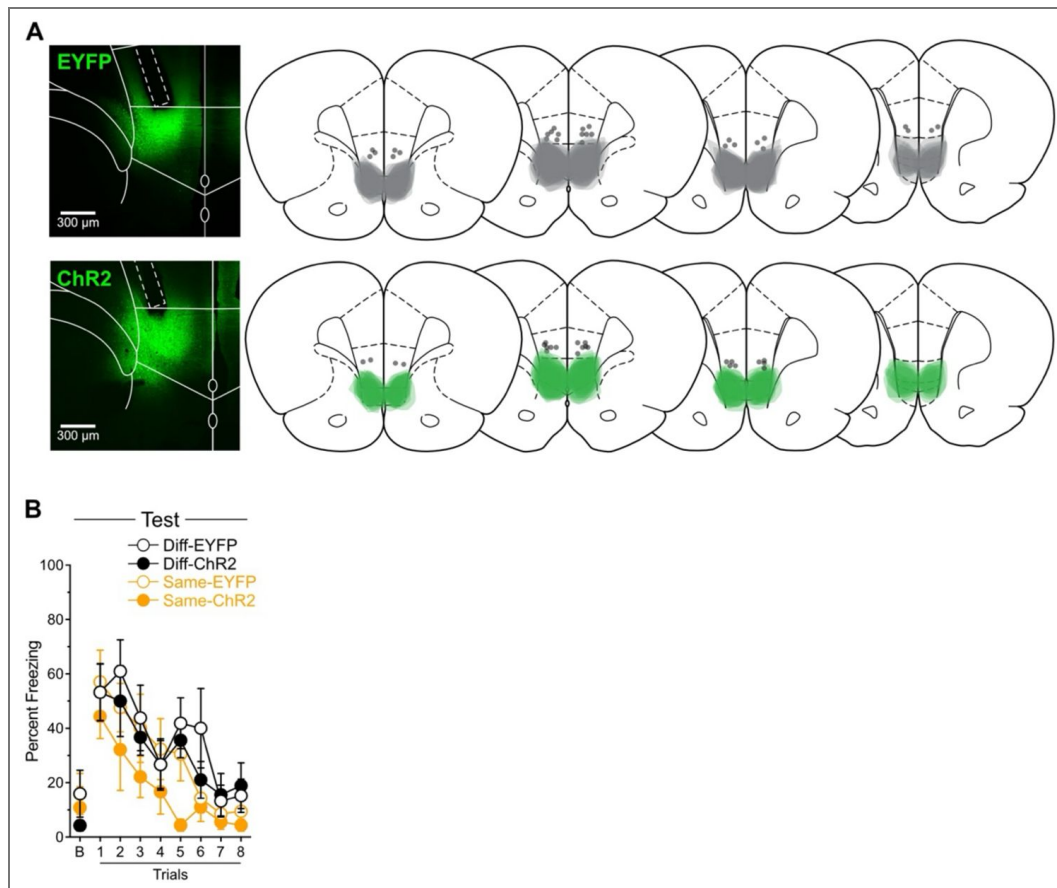


Figure 4 – figure supplemental 1. Histological and behavioral data related to Figure 4B.

A. Data for Experiment 3. Micrographs showing EYFP and ChR2 expression in IL neurons. Minimal (light grey for EYFP and light green for ChR2) and maximal (dark grey for EYFP and dark green for ChR2) extent of IL viral expression. Location of fiber-optic cannulas is also represented (black circles). From left to right: +3.72; +3.24; +3.00; +2.76 mm from bregma. **B.** Freezing across trials during the test presented in Figure 4B. Freezing decreased across trials (Trial: $F_{(1,22)} = 64.00$; $p < 0.001$; $\eta^2 = 0.74$), but this decrease did not interact with the planned orthogonal contrasts reported in the main manuscript (lowest $p = 0.39$). Data are shown as mean $\pm$ SEM.

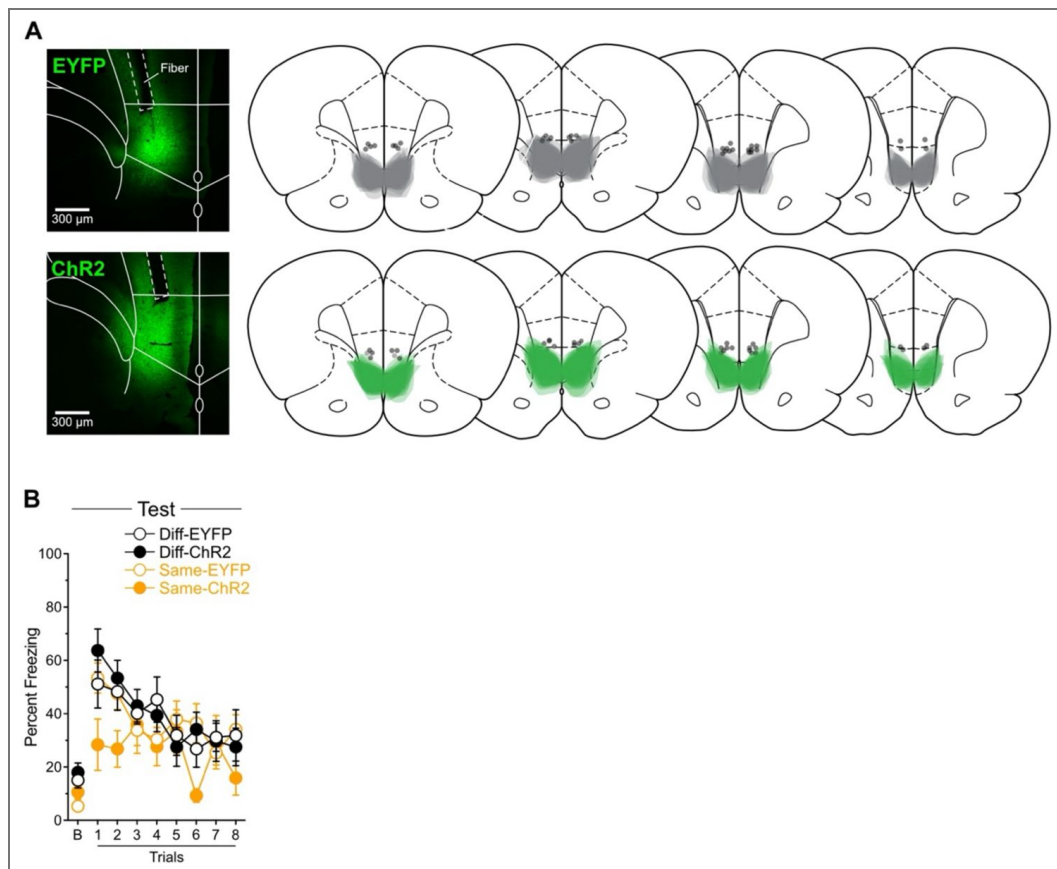


Figure 5 – figure supplemental 1. Histological and behavioral data related to Figure 5.

A. Data for Experiment 5. Micrographs showing EYFP and ChR2 expression in IL neurons. Minimal (light grey for EYFP and light green for ChR2) and maximal (dark grey for EYFP and dark green for ChR2) extent of IL viral expression. Location of fiber-optic cannulas is also represented (black circles). From left to right: +3.72; +3.24; +3.00; +2.76 mm from bregma. **B.** Freezing across trials during the test presented in Figure 5C. Freezing decreased across trials (Trial: $F_{(1,31)} = 25.12$; $p < 0.001$; $\eta^2 = 0.45$), but this decrease did not interact with the planned orthogonal contrasts reported in the main manuscript (lowest $p = 0.23$). Data are shown as mean $\pm$ SEM.

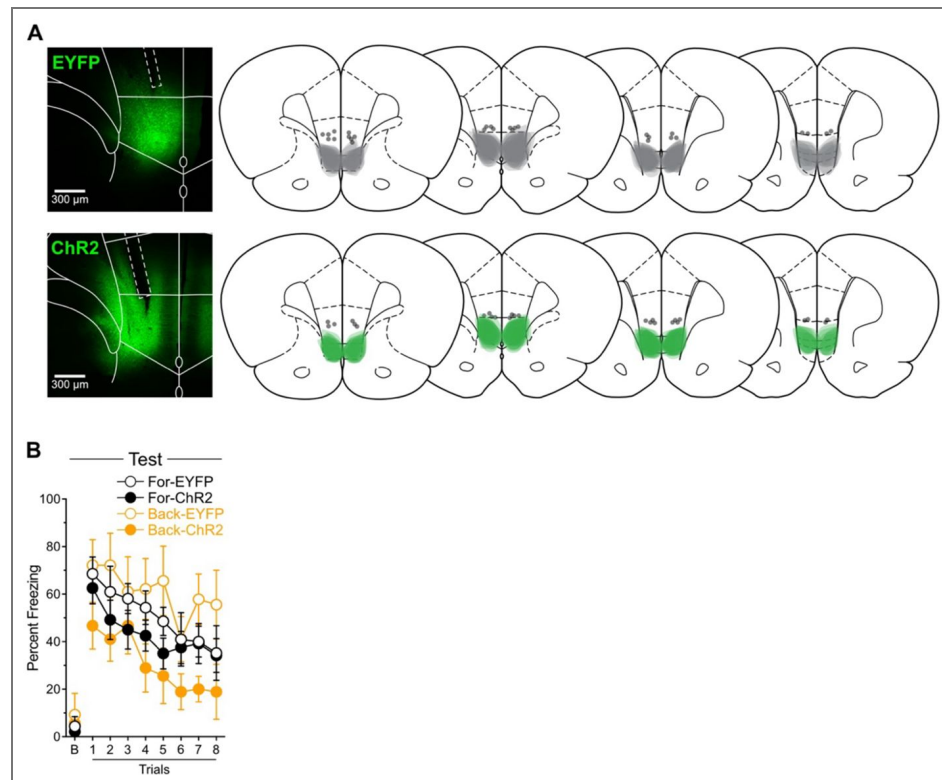


Figure 6 – figure supplemental 1. Histological and behavioral data related to Figure 6.

A. Data for Experiment 6. Micrographs showing EYFP and ChR2 expression in IL neurons. Minimal (light grey for EYFP and light green for ChR2) and maximal (dark grey for EYFP and dark green for ChR2) extent of IL viral expression. Location of fiber-optic cannulas is also represented (black circles). From left to right: +3.72; +3.24; +3.00; +2.76 mm from bregma. **B.** Freezing across trials during the test presented in Figure 6C. Freezing decreased across trials (Trial: $F_{(1,23)} = 52.61$; $p < 0.001$; $\eta^2 = 0.70$), but this decrease did not interact with the planned orthogonal contrasts reported in the main manuscript (lowest $p = 0.34$). Data are shown as mean $\pm$ SEM.

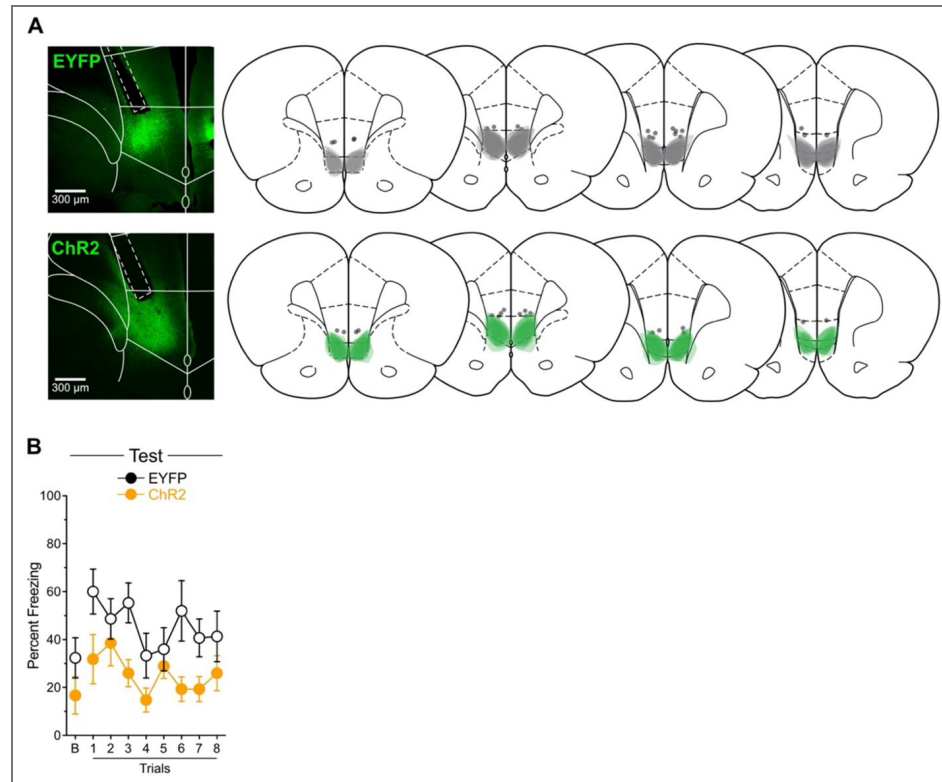


Figure 7 – figure supplemental 1. Histological and behavioral data related to Figure 7C.

A. Data for Experiment 7. Micrographs showing EYFP and ChR2 expression in IL neurons. Minimal (light grey for EYFP and light green for ChR2) and maximal (dark grey for EYFP and dark green for ChR2) extent of IL viral expression. Location of fiber-optic cannulas is also represented (black circles). From left to right: +3.72; +3.24; +3.00; +2.76 mm from bregma. **B.** Freezing across trials during the test presented in Figure 7C. Freezing remained stable across trials (Trial: $p = 0.07$) and the Trial factor did not interact with the planned orthogonal contrast reported in the main manuscript ($p = 0.84$). Data are shown as mean $\pm$ SEM.

Data availability

All data generated and/or analysed during this study are available via the Open Science Framework repository (<https://osf.io/wr7gx/>). The files contain the numerical data used to generate the figures.

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Peer reviews

Reviewer #1 (Public review):

The revised manuscript presents an interesting and technically competent set of experiments exploring the role of the infralimbic cortex (IL) in extinction learning. The inclusion of histological validation in the supplemental material improves the transparency and credibility of the results, and the overall presentation has been clarified. However, several key issues remain that limit the strength of the conclusions.

The behavioral effects reported are modest, as evident from the trial-by-trial data included in the supplemental figures. Although the authors interpret their findings as evidence that IL stimulation facilitates extinction only after prior inhibitory learning, this conclusion is not directly supported by their data. The experiments do not include a condition in which IL stimulation is delivered during extinction training alone, without prior inhibitory experience. Without this control, the claim that prior inhibitory memory is necessary for facilitation remains speculative.

The electrophysiological example provided shows that IL stimulation induces a sustained inhibition that outlasts the stimulation period. This prolonged suppression could potentially interfere with consolidation processes following tone presentation rather than facilitating them. The authors should consider and discuss this alternative interpretation in light of their behavioral data.

It is unfortunate that several animals had to be excluded after histological verification, but the resulting mismatch between groups remains a concern. Without a power analysis indicating the number of subjects required to achieve reliable effects, it is difficult to determine whether the modest behavioral differences reflect genuine biological variability or insufficient statistical power. Additional animals may be needed to properly address this imbalance.

Overall, while the manuscript is improved in clarity and methodological detail, the behavioral effects remain weak, and the mechanistic interpretation requires stronger experimental support and consideration of alternative explanations.

<https://doi.org/10.7554/eLife.108719.2.sa3>

Reviewer #2 (Public review):

Summary:

In this manuscript, the authors examine the mechanisms by which stimulation of the infralimbic cortex (IL) facilitates the retention and retrieval of inhibitory memories. Previous work has shown that optogenetic stimulation of the IL suppresses freezing during extinction but does not improve extinction recall when extinction memory is probed one day later. When stimulation occurs during a second extinction session (following a prior stimulation-free extinction session), freezing is suppressed during the second extinction as well as during the tone test the following day. The current study was designed to further explore the facilitatory role of the IL in inhibitory learning and memory recall. The authors conducted a series of experiments to determine whether recruitment of IL extends to other forms of inhibitory learning (e.g., backward conditioning) and to inhibitory learning involving appetitive conditioning. Further, they assessed whether their effects could be explained by

stimulus familiarity. The results of their experiments show that backward conditioning, another form of inhibitory learning, also enabled IL stimulation to enhance fear extinction. This phenomenon was not specific to aversive learning as backward appetitive conditioning similarly allowed IL stimulation to facilitate extinction of aversive memories. Finally, the authors ruled out the possibility that IL facilitated extinction merely because of prior experience with the stimulus (e.g., reducing the novelty of the stimulus). These findings significantly advance our understanding of the contribution of IL to inhibitory learning. Namely, they show that the IL is recruited during various forms of inhibitory learning and its involvement is independent of the motivational value associated with the unconditioned stimulus.

Strengths to highlight:

- (1) Transparency about the inclusion of both sexes and the representation of data from both sexes in figures.
- (2) Very clear representation of groups and experimental design for each figure.
- (3) The authors were very rigorous in determining the neurobehavioral basis for the effects of IL stimulation on extinction. They considered multiple interpretations and designed experiments to address these possible accounts of their data.
- (4) The rationale for and the design of the experiments in this manuscript are clearly based on a wealth of knowledge about learning theory. The authors leveraged this expertise to narrow down how the IL encodes and retrieves inhibitory memories.

<https://doi.org/10.7554/eLife.108719.2.sa2>

Reviewer #3 (Public review):

Summary:

This is a really nice manuscript with different lines of evidence to show that the IL encodes inhibitory memories that can then be manipulated by optogenetic stimulation of these neurons during extinction. The behavioral designs are excellent, with converging evidence using extinction/re-extinction, backwards/forwards aversive conditioning, and backwards appetitive/forwards aversive conditioning. Additional factors, such as nonassociative effects of the CS or US, also are considered, and the authors evaluate the inhibitory properties of the CS with tests of conditioned inhibition. The authors have addressed the prior reviews. I still think it is unfortunate that the groups were not properly balanced in some of the figures (as noted by the authors, they were matched appropriately in real time, but some animals had to be dropped after histology, which caused some balancing issues). I think the overall pattern of results is compelling enough that more subjects do not need to be added, but it would still be nice to see more acknowledgement and statistical analyses of how these pre-existing differences may have impacted test performance.

Strengths:

The experimental designs are very rigorous with an unusual level of behavioral sophistication.

Weaknesses:

The various group differences in Figure 2 prior to any manipulation are still problematic. There was a reliable effect of subsequent group assignment in Figure 2 ($p < 0.05$, described as "marginal" in multiple places). Then there are differences in extinction (nonsignificant at $p = .07$). The test difference between ReExt OFF/ON is identical to the difference at the end of

extinction and the beginning of Forward 2, in terms of absolute size. I really don't think much can be made of the test result. The authors state in their response that this difference was not evident during the forward phase, but there clearly is a large ordinal difference on the first trial. I think it is appropriate to only focus on test differences when groups are appropriately matched, but when there are pre-existing differences (even when not statistically significant) then they really need to be incorporated into the statistical test somehow.

The same problem is evident in Figure 4B, but here the large differences in the Same groups are opposite to the test differences. It's hard to say how those large differences ultimately impacted the test results. I suppose it is good that the differences during Forward conditioning did not ultimately predict test differences, but this really should have been addressed with more subjects in these experiments. The authors explore the interactions appropriately but with $n=6$ in the various subgroups, it's not surprising that some of these effects were not detected statistically.

It is useful to see the trial-by-trial test data now presented in the supplement. I think the discussion does a good job of addressing the issues of retrieval, but the ideas of Estes about session cues that the authors bring up in their response haven't really held up over the years (e.g., Robbins, 1990, who explicitly tested this; other demonstrations of within-session spontaneous recovery), for what it's worth.

<https://doi.org/10.7554/eLife.108719.2.sa1>

Author response:

The following is the authors' response to the current reviews.

Public Reviews:

Reviewer #1 (Public review):

The revised manuscript presents an interesting and technically competent set of experiments exploring the role of the infralimbic cortex (IL) in extinction learning. The inclusion of histological validation in the supplemental material improves the transparency and credibility of the results, and the overall presentation has been clarified. However, several key issues remain that limit the strength of the conclusions.

We thank the Reviewer for their positive assessment of our revised manuscript. We discussed the issues raised by the Reviewer below.

The behavioral effects reported are modest, as evident from the trial-by-trial data included in the supplemental figures. Although the authors interpret their findings as evidence that IL stimulation facilitates extinction only after prior inhibitory learning, this conclusion is not directly supported by their data. The experiments do not include a condition in which IL stimulation is delivered during extinction training alone, without prior inhibitory experience. Without this control, the claim that prior inhibitory memory is necessary for facilitation remains speculative.

The manuscript provides evidence across five experiments (Figures 2-6) that IL stimulation fails to facilitate extinction training in the absence of prior inhibitory experience. We therefore remain confident that the data support our conclusion: prior inhibitory learning enables IL stimulation to facilitate subsequent inhibitory learning.

The electrophysiological example provided shows that IL stimulation induces a sustained inhibition that outlasts the stimulation period. This prolonged suppression could potentially interfere with consolidation processes following tone presentation rather than

facilitating them. The authors should consider and discuss this alternative interpretation in light of their behavioral data.

The possibility that IL stimulation exerted its effects by interfering with consolidation processes is inconsistent with the literature. Disrupting consolidation processes in the IL impairs extinction learning (1), even when animals have prior inhibitory learning experience (2). Yet our experiments found that IL stimulation failed to interfere with initial extinction learning but instead facilitated subsequent learning. Furthermore, the electrophysiological example demonstrates that the inhibitory effect is transient: the cell returned to firing properties similar to those observed pre-stimulation, making it unlikely that inhibition persists during the consolidation window.

It is unfortunate that several animals had to be excluded after histological verification, but the resulting mismatch between groups remains a concern. Without a power analysis indicating the number of subjects required to achieve reliable effects, it is difficult to determine whether the modest behavioral differences reflect genuine biological variability or insufficient statistical power. Additional animals may be needed to properly address this imbalance.

As noted in the revised manuscript, we are confident about the reliability of the findings reported. The manuscript provides evidence across five experiments that IL stimulation fails to facilitate brief extinction in the absence of prior inhibitory experience, replicating previous findings (3, 4). The manuscript also replicates these prior studies by demonstrating that experience with either fear or appetitive extinction enables IL stimulation to facilitate subsequent fear extinction. Furthermore, the present experiments replicate the facilitative effects of IL stimulation following fear or appetitive backward conditioning.

Overall, while the manuscript is improved in clarity and methodological detail, the behavioral effects remain weak, and the mechanistic interpretation requires stronger experimental support and consideration of alternative explanations.

We respectfully disagree with the assertion that the reported results are weak. The manuscript replicates all main findings internally or reproduces findings from previously published studies. While alternative explanations cannot be entirely excluded, we are not aware of any competing account that predicts the pattern of results reported here.

Reviewer #2 (Public review):

Summary:

In this manuscript, the authors examine the mechanisms by which stimulation of the infralimbic cortex (IL) facilitates the retention and retrieval of inhibitory memories. Previous work has shown that optogenetic stimulation of the IL suppresses freezing during extinction but does not improve extinction recall when extinction memory is probed one day later. When stimulation occurs during a second extinction session (following a prior stimulation-free extinction session), freezing is suppressed during the second extinction as well as during the tone test the following day. The current study was designed to further explore the facilitatory role of the IL in inhibitory learning and memory recall. The authors conducted a series of experiments to determine whether recruitment of IL extends to other forms of inhibitory learning (e.g., backward conditioning) and to inhibitory learning involving appetitive conditioning. Further, they assessed whether their effects could be explained by stimulus familiarity. The results of their experiments show that backward conditioning, another form of inhibitory learning, also enabled IL stimulation to enhance fear extinction. This phenomenon was not specific to aversive learning as backward appetitive conditioning similarly allowed IL stimulation to facilitate extinction of aversive memories. Finally, the authors ruled out the possibility that IL facilitated extinction merely because of prior experience with the stimulus (e.g.,

reducing the novelty of the stimulus). These findings significantly advance our understanding of the contribution of IL to inhibitory learning. Namely, they show that the IL is recruited during various forms of inhibitory learning and its involvement is independent of the motivational value associated with the unconditioned stimulus.

We thank the Reviewer for their positive assessment.

Strengths to highlight:

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We thank the Reviewer for their positive assessment of our revised manuscript. We discussed the comments regarding group balancing below.

Strengths:

The experimental designs are very rigorous with an unusual level of behavioral sophistication.

We thank the Reviewer for their positive assessment

Weaknesses:

The various group differences in Figure 2 prior to any manipulation are still problematic. There was a reliable effect of subsequent group assignment in Figure 2 ($p < 0.05$, described as "marginal" in multiple places). Then there are differences in extinction (nonsignificant at $p = .07$). The test difference between ReExt OFF/ON is identical to the difference at the end of extinction and the beginning of Forward 2, in terms of absolute size. I really don't think much can be made of the test result. The authors state in their response that this difference was not evident during the forward phase, but there clearly is a large ordinal difference on the first trial. I think it is appropriate to only focus on test differences when groups are appropriately matched, but when there are pre-existing differences (even when not statistically significant) then they really need to be incorporated into the statistical test somehow.

We carefully considered the Reviewer's suggestion, but it is not possible to adjust the statistical analyses at test because these analyses do not directly compare the two ReExt groups. Any scaling of performance would require including the two Ext groups, which is not feasible since these groups did not receive initial extinction. Moreover, the analyses provide no conclusive evidence of pre-existing differences between the two ReExt groups: the difference was not significant during initial extinction and was absent during the Forward 2 stage. We acknowledge that closer performance between the two ReExt groups during initial extinction would have been preferable. However, we remain confident in the results obtained because they replicate previous experiments in which the two ReExt groups displayed identical performance during initial extinction.

The same problem is evident in Figure 4B, but here the large differences in the Same groups are opposite to the test differences. It's hard to say how those large differences ultimately impacted the test results. I suppose it is good that the differences during Forward conditioning did not ultimately predict test differences, but this really should have been addressed with more subjects in these experiments. The authors explore the interactions appropriately but with $n = 6$ in the various subgroups, it's not surprising that some of these effects were not detected statistically.

As the Reviewer noted, the unexpected differences in Figure 4B are opposite in direction to the test differences. Importantly, Figure 4B replicates the main findings from Figure 3, which did not show these unexpected differences.

It is useful to see the trial-by-trial test data now presented in the supplement. I think the discussion does a good job of addressing the issues of retrieval, but the ideas of Estes about session cues that the authors bring up in their response haven't really held up over the years (e.g., Robbins, 1990, who explicitly tested this; other demonstrations of within-session spontaneous recovery), for what it's worth.

We thank the Reviewer for bringing our attention to Robbins' work on session cues. We understand that the issue of retrieval is important but as we noted before, our manuscript and its conclusions do not claim to differentiate retrieval from additional learning.

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The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary:

The manuscript reports a series of experiments designed to test whether optogenetic activation of infralimbic (IL) neurons facilitates extinction retrieval and whether this depends on animals' prior experience. In Experiment 1, rats underwent fear conditioning followed by either one or two extinction sessions, with IL stimulation given during the second extinction; stimulation facilitated extinction retrieval only in rats with prior extinction experience. Experiments 2 and 3 examined whether backward conditioning (CS presented after the US) could establish inhibitory properties that allowed IL stimulation to enhance extinction, and whether this effect was specific to the same stimulus or generalized to different stimuli. Experiments 5 - 7 extended this approach to appetitive learning: rats received backward or forward appetitive conditioning followed by extinction, and then fear conditioning, to determine whether IL stimulation could enhance extinction in contexts beyond aversive learning and across conditioning sequences. Across studies, the key claim is that IL activation facilitates extinction retrieval only when animals possess a prior inhibitory memory, and that this effect generalizes across aversive and appetitive paradigms.

Strengths:

(1) The design attempts to dissect the role of IL activity as a function of prior learning, which is conceptually valuable.

We thank the Reviewer for their positive assessment.

(2) The experimental design of probing different inhibitory learning approaches to probe how IL activation facilitates extinction learning was creative and innovative.

We thank the Reviewer for their positive assessment.

Weaknesses:

(1) Non-specific manipulation.

ChR2 was expressed in IL without distinction between glutamatergic and GABAergic populations. Without knowing the relative contribution of these cell types or the percentage of neurons affected, the circuit-level interpretation of the results is unclear.

ChR2 was intentionally expressed in the infralimbic cortex (IL) without distinction between local neuronal populations for two reasons. First, the primary aim of this was to uncover some of the features characterizing the encoding of inhibitory memories in the IL, and this encoding likely engages interactions among various neuronal populations within the IL. Second, the hypotheses tested in the manuscript derived from findings that indiscriminately stimulated the IL using the GABA_A receptor antagonist picrotoxin, which is best mimicked by the approach taken. We agree that it is also important to determine the respective

contributions of distinct IL neuronal populations to inhibitory encoding; however, the global approach implemented in the present experiments represents a necessary initial step. These matters have been incorporated in the Discussion of the revised manuscript.

(2) Extinction retrieval test conflates processes

The retrieval test included 8 tones. Averaging across this many tone presentations conflate extinction retrieval/expression (early tones) with further extinction learning (later tones). A more appropriate analysis would focus on the first 2-4 tones to capture retrieval only. As currently presented, the data do not isolate extinction retrieval.

It is unclear when retrieval of what has been learned across extinction ceases and additional extinction learning occurs. In fact, it is only the first stimulus presentation that unequivocally permits a distinction between retrieval and additional extinction learning, as the conditions for this additional learning have not been fulfilled at that presentation. However, confining evidence for retrieval to the first stimulus presentation introduces concerns that other factors could influence performance. For instance, processing of the stimulus present at the start of the session may differ from that present at the end of the previous session, thereby affecting what is retrieved. Such differences between the stimuli present at the start and end of an extinction session have been long recognized as a potential explanation for spontaneous recovery (Estes, 1955). More importantly, whether the test data presented confound retrieval and additional extinction learning or not, the interpretation remains the same with respect to the effects of a prior history of inhibitory learning on enabling the facilitative effects of IL stimulation. Finally, it is unclear how these facilitative effects could occur in the absence of the subjects retrieving the extinction memory formed under the stimulation. Nevertheless, the revised manuscript now provides the trial-by-trial performance (see Supplemental Figure 3) during the post-extinction retrieval tests and addresses this issue in the Discussion.

(3) Under-sampling and poor group matching.

Sample sizes appear small, which may explain why groups are not well matched in several figures (e.g., 2b, 3b, 6b, 6c) and why there are several instances of unexpected interactions (protocol, virus, and period). This baseline mismatch raises concerns about the reliability of group differences.

Efforts were made to match group performance upon completion of each training stage and before IL stimulation. Unfortunately, these efforts were not completely successful due to exclusions following post-mortem analyses. This has been made explicit in the revised manuscript (Materials and Methods, Subjects section). However, we acknowledge that the unexpected interactions deserve further discussion, and this has been incorporated into the revised manuscript (see also comment from Reviewer 2). Although we cannot exclude the possibility that sample sizes may have contributed to some of these interactions, we remain confident about the reliability of the main findings reported, especially given their replication across the various protocols. Overall, the manuscript provides evidence that IL stimulation does not facilitate brief extinction in the absence of prior inhibitory experience in five different experiments, replicating previous findings (Lingawi et al., 2018; Lingawi et al., 2017). It also replicates these previous findings by showing that prior experience with either fear or appetitive extinction enables IL stimulation to facilitate subsequent fear extinction. Furthermore, the facilitative effects of such stimulation following fear or appetitive backward conditioning are replicated in the present manuscript. This is discussed in the Discussion of the revised manuscript.

(4) Incomplete presentation of conditioning data

Figure 3 only shows a single conditioning session despite five days of training. Without the full dataset, it is difficult to evaluate learning dynamics or whether groups were equivalent before testing.

We apologize, as we incorrectly labeled the X axis for the backward conditioning data in Figures 3B, 4B, 4D and 5B. It should have indicated “Days” instead of “Trials”. This error has been corrected in the revised manuscript (see also second comment from Reviewer 2).

(5) Interpretation stronger than evidence.

The authors conclude that IL activation facilitates extinction retrieval only when an inhibitory memory has been formed. However, given the caveats above, the data are insufficient to support such a strong mechanistic claim. The results could reflect nonspecific facilitation or disruption of behavior by broad prefrontal activation. Moreover, there is compelling evidence that optogenetic activation of IL during fear extinction does facilitate subsequent extinction retrieval without prior extinction training (DoMonte et al 2015, Chen et al 2021), which the authors do not directly test in this study.

As noted above, the interpretations of the main findings stand whether the test data confounds retrieval with additional extinction learning or not. The revised manuscript also clarifies the plotting of the data for the backward conditioning stages. We do agree that further discussion of the unexpected interactions is necessary, and this has been incorporated into the revised manuscript. However, the various replications of the core findings provide strong evidence for their reliability and the interpretations advanced in the original manuscript. The proposal that the results reflect non-specific facilitation or disruption of behavior seems highly unlikely. Indeed, the present experiments and previous findings (Lingawi et al., 2018; Lingawi et al., 2017) provide multiple demonstrations that IL stimulation fails to produce any facilitation in the absence of prior inhibitory experience with the target stimulus. Although these demonstrations appear inconsistent with previous studies (Do-Monte et al., 2015; Chen et al., 2021), this inconsistency is likely explained by the fact that these studies manipulated activity in specific IL neuronal populations. Previous work has already revealed differences between manipulations targeting discrete IL neuronal populations as opposed to general IL activity (Kim et al., 2016). Importantly, as previously noted, the present manuscript aimed to generally explore inhibitory encoding in the IL that is likely to engage several neuronal populations within the IL. Adequate statements on these matters have been included in the Discussion of the revised manuscript.

Reviewer #2 (Public review):

Summary:

In this manuscript, the authors examine the mechanisms by which stimulation of the infralimbic cortex (IL) facilitates the retention and retrieval of inhibitory memories. Previous work has shown that optogenetic stimulation of the IL suppresses freezing during extinction but does not improve extinction recall when extinction memory is probed one day later. When stimulation occurs during a second extinction session (following a prior stimulation-free extinction session), freezing is suppressed during the second extinction as well as during the tone test the following day. The current study was designed to further explore the facilitatory role of the IL in inhibitory learning and memory recall. The authors conducted a series of experiments to determine whether recruitment of IL extends to other forms of inhibitory learning (e.g., backward conditioning) and to inhibitory learning involving appetitive conditioning. Further, they assessed whether their effects could be explained by stimulus familiarity. The results of their experiments show that backward conditioning, another form of inhibitory learning, also enabled IL stimulation to enhance fear extinction. This phenomenon was not specific to aversive learning, as backward appetitive conditioning similarly allowed IL stimulation to facilitate extinction of aversive memories. Finally, the authors ruled out the possibility that IL facilitated extinction merely because of prior experience with the stimulus (e.g., reducing the novelty of the stimulus). These findings significantly advance our

understanding of the contribution of IL to inhibitory learning. Namely, they show that the IL is recruited during various forms of inhibitory learning, and its involvement is independent of the motivational value associated with the unconditioned stimulus.

Strengths:

(1) Transparency about the inclusion of both sexes and the representation of data from both sexes in figures.

We thank the Reviewer for their positive assessment.

(2) Very clear representation of groups and experimental design for each figure.

We thank the Reviewer for their positive assessment.

(3) The authors were very rigorous in determining the neurobehavioral basis for the effects of IL stimulation on extinction. They considered multiple interpretations and designed experiments to address these possible accounts of their data.

We thank the Reviewer for their positive assessment.

(4) The rationale for and the design of the experiments in this manuscript are clearly based on a wealth of knowledge about learning theory. The authors leveraged this expertise to narrow down how the IL encodes and retrieves inhibitory memories.

We thank the Reviewer for their positive assessment.

Weaknesses:

(1) In Experiment 1, although not statistically significant, it does appear as though the stimulation groups (OFF and ON) differ during Extinction 1. It seems like this may be due to a difference between these groups after the first forward conditioning. Could the authors have prevented this potential group difference in Extinction 1 by re-balancing group assignment after the first forward conditioning session to minimize the differences in fear acquisition (the authors do report a marginally significant effect between the groups that would undergo one vs. two extinction sessions in their freezing during the first conditioning session)?

Efforts were made daily to match group performance across the training stages, but these efforts were ultimately hampered by the necessary exclusions following postmortem analyses. This has been made explicit in the revised manuscript (Materials and Methods, Subjects section). Regarding freezing during Extinction 1, as noted by the Reviewer, the difference, which was not statistically significant, was absent across trials during the subsequent forward fear conditioning stage. Likewise, the protocol difference observed during the initial forward fear conditioning was absent in subsequent stages. We are therefore confident that these initial differences (significant or not) did not impact the main findings at test. Importantly, these findings replicate previous work using identical protocols in which no differences were present during the training stages. These considerations have been addressed in the revised manuscript (see Results for Experiment 1).

(2) Across all experiments (except for Experiment 1), the authors state that freezing during the initial conditioning increased across "days". The figures that correspond to this text, however, show that freezing changes across trials. In the methods, the authors report that backward conditioning occurred over 5 days. It would be helpful to understand how these data were analyzed and collated to create the final figures. Was the freezing averaged across the five days for each trial for analyses and figures?

We apologize, as noted above, for having incorrectly labeled the X axis across the backward conditioning data sets in Figures 3B, 4B, 4D and 5B. It should have indicated “Days” instead of “Trials”. The data shown in these Figures use the average of all trials on a given day. This has been clarified in the methods section of the revised manuscript (Statistical Analyses section). The labeling errors on the Figures have been corrected.

(3) In Experiment 3, the authors report a significant Protocol X Virus interaction. It would be useful if the authors could conduct post-hoc analyses to determine the source of this interaction. Inspection of Figure 4B suggests that freezing during the two different variants of backward conditioning differs between the virus groups. Did the authors expect to see a difference in backward conditioning depending on the stimulus used in the conditioning procedure (light vs. tone)? The authors don't really address this confounding interaction, but I do think a discussion is warranted.

We agree with the Reviewer that further discussion of the Protocol x Virus interaction that emerged during the backward conditioning and forward conditioning stages of Experiment 3 is warranted. This discussion has been provided in the revised manuscript (see Results section). Briefly, during both stages, follow-up analyses did not reveal any differences (main effects or interactions) between the two groups trained with the light stimulus (Diff-EYFP and Diff-ChR2). By contrast, the ChR2 group trained with the tone (Back-ChR2) froze more overall than the EYFP group (Back-EYFP), but there were no other significant differences between the two groups. Based on these analyses, the Protocol x Virus interaction appears to be driven by greater freezing in the ChR2 group trained with the tone rather than a difference in the backward conditioning performance based on stimulus identity. Consistent with this, the statistical analyses did not reveal a main effect of Protocol during either the backward conditioning stage or the stimulus trials during the forward conditioning stage. Nevertheless, during this latter stage, a main effect of Protocol emerged during baseline performance, but once again, this seems to be driven by the Back-ChR2 group. Critically, it is unclear how greater stimulus freezing in the Back-ChR2 group during forward conditioning would lead to lower freezing during the post-extinction retrieval test.

We note that an unexpected Protocol x Period interaction was found during appetitive backward conditioning in Experiment 5. For consistency, we conducted additional analyses to determine the source of this interaction (see Results section). As previously noted, performance during appetitive backward conditioning is noisy and cannot be taken as a failure to generate inhibitory learning. It is therefore unlikely that this interaction implied a difference in such learning.

(4) In this same experiment, the authors state that freezing decreased during extinction; however, freezing in the Diff-EYFP group at the start of extinction (first bin of trials) doesn't look appreciably different than their freezing at the end of the session. Did this group actually extinguish their fear? Freezing on the tone test day also does not look too different from freezing during the last block of extinction trials.

We confirm that overall, there was a significant decline in freezing across the extinction session shown in Figure 4B. The Reviewer is correct to point out that this decline was modest (if not negligible) in the Diff-EYFP group, which was receiving its first inhibitory training with the target tone stimulus. It is worth noting that across all experiments, most groups that did not receive infralimbic stimulation displayed a modest decline in freezing during the extinction session since it was relatively brief, involving only 6 or 8 tone alone presentations. This was intentional, as we aimed for the brief extinction session to generate minimal inhibitory learning and thereby to detect any facilitatory effect of infralimbic stimulation. This has been clarified and explained in the revised version of the manuscript (see Results section, description of Experiment 1).

(5) The Discussion explored the outcomes of the experiments in detail, but it would be useful for the authors to discuss the implications of their findings for our understanding of circuits in which the IL is embedded that are involved in inhibitory learning and memory. It would also be useful for the authors to acknowledge in the Discussion that although they did not have the statistical power to detect sex differences, future work is needed to explore whether IL functions similarly in both sexes.

In line with the Reviewer's suggestion (see also Reviewer 3), the Discussion section has been substantially altered in the revised manuscript. Among other things, it does mention that future studies will need to examine the role of additional brain regions in the effects reported and it acknowledges the need to further explore sex differences and IL functions.

Reviewer #3 (Public review):

Summary:

This is a really nice manuscript with different lines of evidence to show that the IL encodes inhibitory memories that can then be manipulated by optogenetic stimulation of these neurons during extinction. The behavioral designs are excellent, with converging evidence using extinction/re-extinction, backwards/forwards aversive conditioning, and backwards appetitive/forwards aversive conditioning. Additional factors, such as nonassociative effects of the CS or US, are also considered, and the authors evaluate the inhibitory properties of the CS with tests of conditioned inhibition.

Strengths:

The experimental designs are very rigorous with an unusual level of behavioral sophistication.

We thank the Reviewer for their positive assessment

Weaknesses:

(1) More justification for parametric choices (number of days of backwards vs forwards conditioning) could be provided.

All experimental parameters were based on previously published experiments showing the capacity of the backward conditioning protocols to generate inhibitory learning and the forward conditioning protocols to produce excitatory learning. Although this was mentioned in the methods section, we acknowledge that further explanation was required to justify the need for multiple days of backward training. This has been provided in the revised manuscript (see Results section and description of the backward parameters).

(2) The current discussion could be condensed and could focus on broader implications for the literature.

The discussion has been severely condensed and broader implications have been discussed with respect to the existing literature looking at the neural circuitry underlying inhibitory learning.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

(1) Re-analyze extinction retrieval, focusing only on the first 2-4 tones to capture extinction expression.

This recommendation corresponds to the second public comment made by the Reviewer, and we have replied to this comment.

(2) Directly test whether activation of IL during fear extinction is insufficient to facilitate extinction retrieval without prior extinction training.

The manuscript provides five separate demonstrations that the optogenetic approach to stimulate IL activity did not facilitate the initial brief extinction session. This reproduces what had been found with indiscriminate pharmacological stimulation in our previous research (Lingawi et al., 2018; Lingawi et al., 2017). We appreciate that other work that stimulated specific IL neuronal populations has observed facilitation of extinction but, the present manuscript focuses on the role of all IL neuronal populations in encoding inhibitory memories. The Reviewer's request would imply contrasting the role of various neuronal populations, which is beyond the scope of this manuscript. Nevertheless, we have modified our discussion to indicate that future research should establish which IL neuronal population(s) contribute to the effects reported here.

(3) Show the percentage of neurons that exhibit excitatory or inhibitory responses in IL after non-specific optogenetic activation to better understand how this manipulation is affecting IL circuitry.

All electrophysiological recordings (n = 10 cells) are presented in Figure 1C. Chr2 excitation was substantial and overwhelming. Based on the physiological and morphological characteristics of the recorded cells, one was non-pyramidal and was excited by LED light delivery. The remaining 9 cells were pyramidal. One did not respond to LED delivery, but we cannot exclude the possibility that this was due to a lack of Chr2 expression in the somatic compartment. Another cell showed a mild reduction in activity following LED stimulation, while the remaining 7 cells displayed clear excitation upon LED stimulation. We have modified our manuscript to reflect these observations. We did not include percentages since only 10 recordings are shown.

(4) Present data from all five conditioning sessions, not just one, to allow evaluation of learning history.

This recommendation corresponds to the fourth public comment made by the Reviewer, and we have replied to this comment.

(5) Address the issue of small and poorly matched groups, particularly in Figures 2b, 3b, 6b, and 6c.

This recommendation corresponds to the third public comment made by the Reviewer, and we have replied to this comment.

(6) Temper the conclusions to reflect the limitations of sampling, group matching, and the lack of specificity in the manipulation.

We have modified our Discussion to address potential issues related to sampling and group matching. However, we are unsure how the lack of specificity of the IL stimulation has any impact on the interpretations made, since no statement is made about neuronal specificity. That said, as noted above, "we have modified our discussion to indicate that future research should establish which IL neuronal population(s) contribute to the effects reported here".

Reviewer #2 (Recommendations for the authors):

Nothing additional to include beyond what is written for public view.

Reviewer #3 (Recommendations for the authors):

This is a really nice manuscript with different lines of evidence to show that the IL encodes inhibitory memories that can then be manipulated by optogenetic stimulation of these neurons during extinction. The behavioral designs are excellent, with converging evidence using extinction/re-extinction, backwards/forwards aversive conditioning, and backwards appetitive/forwards aversive conditioning. Additional factors, such as nonassociative effects of the CS or US, are also considered, and the authors evaluate the inhibitory properties of the CS with tests of conditioned inhibition. I only have a couple of comments that the authors may want to consider.

We thank the Reviewer for their positive assessment.

First, in Figure 2, it is unfortunate that there is a general effect of the LED assignment before the LED experience ($p=.07$ during that first extinction session). This is in the same direction as the difference during the test, so it is not clear that the test difference really reflects differences due to Extinction 2 treatment or to preexisting differences based on group assignments.

The Reviewer's comment is identical to the first public comment of Reviewer 2, which has been addressed.

Second, it is notable that the backwards fear conditioning phase was conducted over 5 days, but the forward conditioning phase was conducted over one day. The rationale for these differences should be presented. There is an old idea going back to Konorski that backwards conditioning may lead to excitation initially, and it is only after more extensive trials that inhibitory conditioning occurs (a finding supported by Heth, 1976). Some discussion of the potential biphasic nature of backwards conditioning would be useful, especially for people who want to run this type of experiment but with only a single session of backwards conditioning.

In line with the Reviewer's suggestion, the revised manuscript (see results section) provide an explanation for conducting backward conditioning across multiple days.

Third, as written, each paragraph of the discussion is mostly a recapitulation of the findings from each experiment. This could be condensed significantly, and it would be nice to see more integration with the current literature and how these results challenge or suggest nuance in current thinking about IL function.

We have significantly condensed the recapitulation of our findings in the Discussion of the revised manuscript. The Discussion now dedicates space to address comments from the other Reviewers and integrate the present findings with the current literature.

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